Activité de redémarrage

1.Teste d'accès SSH à la VM à partir la machine physique :

```
arij@server26:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
     Active: active (running) since Tue 2026-09-29 21:12:39 UTC; 8min ago
 Invocation: 3d1599629ba34fe2b2cb15167555de4d
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 1869 (sshd)
      Tasks: 1 (limit: 1426)
     Memory: 5.2M (peak: 7.1M)
        CPU: 145ms
     CGroup: /system.slice/ssh.service
             └─1869 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

sept. 29 21:12:39 server26 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
sept. 29 21:12:39 server26 sshd[1869]: Server listening on 0.0.0.0 port 22.
sept. 29 21:12:39 server26 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
sept. 29 21:12:39 server26 sshd[1869]: Server listening on :: port 22.
sept. 29 21:12:41 server26 sshd-session[1871]: Accepted password for arij from 192.168.159.1 port 58970 ssh2
sept. 29 21:12:41 server26 sshd-session[1871]: pam_unix(sshd:session): session opened for user arij(uid=1000) by arij(uid=0)
arij@server26:~$ ss -tlnp | grep :22
LISTEN 0      4096         0.0.0.0:22       0.0.0.0:*
LISTEN 0      4096            [::]:22          [::]:*
arij@server26:~$ _
```

2. **Sécurisation du SSH :**

```
C:\Users\arijb>ssh-keygen -t ed25519 -C "arij@server26"
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\arijb/.ssh/id_ed25519):
C:\Users\arijb/.ssh/id_ed25519 already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\arijb/.ssh/id_ed25519
Your public key has been saved in C:\Users\arijb/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:A756EGgyM06KnqLCYyvCEcZVT8apxUd8xczcGxjG/ZY arij@server26
The key's randomart image is:
+--[ED25519 256]--+
|     ..oo+.  B+=  |
|    .  += o ..B + |
|. ..  +.. .     =|
|==o .o .       Eo|
|**.  .. S      . |
|oo  .  . .       |
|+ o  ..          |
|=B   ..          |
|O.o ..           |
+----[SHA256]-----+

C:\Users\arijb>
```

3.Puis Lancement la commande de copie et teste

```
C:\Users\arijb>type %USERPROFILE%\.ssh\id_ed25519.pub | ssh arij@192.168.159.144 "mkdir -p ~/.ssh && chmod 700 ~/.ssh && cat >> ~/.ss
h/authorized_keys && chmod 600 ~/.ssh/authorized_keys"

C:\Users\arijb>ssh arij@192.168.159.144
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of mar. 29 sept 2026 21:32:09 UTC

  System load:  0.03               Processes:             226
  Usage of /:   28.9% of 23.45GB   Users logged in:       0
  Memory usage: 12%                IPv4 address for ens33: 192.168.159.144
  Swap usage:   0%


La maintenance de sécurité étendue pour Applications n'est pas activée.

134 mises à jour peuvent être appliquées immédiatement.
113 de ces mises à jour sont des mises à jour de sécurité.
Pour afficher ces mises à jour supplémentaires, exécuter : apt list --upgradable

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez https://ubuntu.com/esm ou executez : sudo pro status


Last login: Tue Sep 29 21:24:30 2026 from 192.168.159.1
arij@server26:~$
```

4. installer Docker

a) Dépendances :

```
arij@server26:~$ sudo apt install ca-certificates curl -y
ca-certificates est déjà la version la plus récente (20260601~26.04.1).
ca-certificates passé en « installé manuellement ».
curl est déjà la version la plus récente (8.18.0-1ubuntu2.7).
curl passé en « installé manuellement ».
Sommaire :
  Mise à niveau de : 0. Installation de : 0, Supprimé : 0. Non mis à jour : 0
arij@server26:~$
```

b) Clé GPG Docker :

```
arij@server26:~$ sudo install -m 0755 -d /etc/apt/keyrings
arij@server26:~$ sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
arij@server26:~$ sudo chmod a+r /etc/apt/keyrings/docker.asc
arij@server26:~$
```

c) Ajout du dépôt Docker :

```
arij@server26:~$ echo \
  "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubun
tu \
  $(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
  sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
arij@server26:~$ cat /etc/apt/sources.list.d/docker.list
deb [arch=amd64 signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu   resolute stable
```

d) Mise à jour + installation de Docker :

```
arij@server26:~$ sudo apt update
Atteint : 1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Atteint : 2 http://tn.archive.ubuntu.com/ubuntu resolute InRelease
Atteint : 3 http://tn.archive.ubuntu.com/ubuntu resolute-updates InRelease
Réception de : 4 https://download.docker.com/linux/ubuntu resolute InRelease [32,5 kB]
Atteint : 5 http://tn.archive.ubuntu.com/ubuntu resolute-backports InRelease
Réception de : 6 https://download.docker.com/linux/ubuntu resolute/stable amd64 Packages [26,8 kB]
59,3 ko réceptionnés en 1s (118 ko/s)
Tous les paquets sont à jour.
```

```
arij@server26:~$ sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin -y
Installation de :
  containerd.io  docker-buildx-plugin  docker-ce  docker-ce-cli  docker-compose-plugin

Installation de dépendances :
  docker-ce-rootless-extras  pigz

Paquets suggérés :
  cgroupfs-mount  | cgroup-lite  docker-model-plugin

Sommaire :
  Mise à niveau de : 0. Installation de : 7, Supprimé : 0. Non mis à jour : 0
Taille du téléchargement : 101 Mo
  Espace nécessaire : 383 Mo / 15,9 Go disponible

Réception de : 1 https://download.docker.com/linux/ubuntu resolute/stable amd64 containerd.io amd64 2.3.6-1~ubuntu.26.04
~resolute [23,2 MB]
Réception de : 2 http://tn.archive.ubuntu.com/ubuntu resolute/universe amd64 pigz amd64 2.8-1build1 [66,8 kB]
Réception de : 3 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-ce-cli amd64 5:29.8.1-1~ubuntu.26
.04~resolute [17,6 MB]
Réception de : 4 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-ce amd64 5:29.8.1-1~ubuntu.26.04~
resolute [24,3 MB]
Réception de : 5 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-buildx-plugin amd64 0.37.1-1~ubun
tu.26.04~resolute [17,3 MB]
Réception de : 6 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-ce-rootless-extras amd64 5:29.8.1
-1~ubuntu.26.04~resolute [10,2 MB]
Réception de : 7 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-compose-plugin amd64 5.5.1-1~ubun
tu.26.04~resolute [8 013 kB]
101 Mo réceptionnés en 20s (5 112 ko/s)
```

e) Vérifier que le service tourne :

```
arij@server26:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-09-29 21:58:59 UTC; 1min 46s ago
 Invocation: fe469debe3654fc4a009f8f832223dd5
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 2543 (dockerd)
      Tasks: 9
     Memory: 31.3M (peak: 31.6M)
        CPU: 1.057s
     CGroup: /system.slice/docker.service
             └─2543 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

sept. 29 21:58:58 server26 dockerd[2543]: time="2026-09-29T21:58:58.122536925Z" level=info msg="Restoring containers: s>
sept. 29 21:58:58 server26 dockerd[2543]: time="2026-09-29T21:58:58.184067911Z" level=info msg="Deleting nftables IPv4 >
```

f) Autorisation à utiliser Docker sans sudo :

```
arij@server26:~$ sudo usermod -aG docker $USER
arij@server26:~$
```

g) Se déconnecter puis se reconnecter :

```
arij@server26:~$ exit
logout
Connection to 192.168.159.144 closed.

C:\Users\arijb>ssh arij@192.168.159.144
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-34-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of mar. 29 sept 2026 22:03:14 UTC

  System load:  0.0               Processes:              225
  Usage of /:   33.3% of 23.45GB  Users logged in:        0
  Memory usage: 14%               IPv4 address for ens33: 192.168.159.144
  Swap usage:   0%


La maintenance de sécurité étendue pour Applications n'est pas activée.

0 mise à jour peut être appliquée immédiatement.

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez https://ubuntu.com/esm ou executez : sudo pro status


Last login: Tue Sep 29 21:52:18 2026 from 192.168.159.1
```

h) Test final :

```
arij@server26:~$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:5e23090353324d887c48ad5e5c56d294eab81588df9605b07d1afe895f9cc8f8
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

5.la finalisation de la sécurisation :

```
arij@server26:~$ sudo nano /etc/ssh/sshd_config.d/00-hardening.conf
[sudo: authenticate] Password:
arij@server26:~$ sudo systemctl reload ssh
sudo sshd -T | grep -E "passwordauthentication|permitrootlogin|pubkeyauthentication"
permitrootlogin no
pubkeyauthentication yes
passwordauthentication no
arij@server26:~$
```

```
C:\Users\arijb>ssh arij@192.168.159.144
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-34-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of mar. 29 sept 2026 22:13:07 UTC

  System load:  0.0               Processes:             225
  Usage of /:   33.3% of 23.45GB  Users logged in:       0
  Memory usage: 14%               IPv4 address for ens33: 192.168.159.144
  Swap usage:   0%


La maintenance de sécurité étendue pour Applications n'est pas activée.

0 mise à jour peut être appliquée immédiatement.

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez https://ubuntu.com/esm ou executez : sudo pro status


Last login: Tue Sep 29 22:11:20 2026 from 192.168.159.1
arij@server26:~$ ssh -o PreferredAuthentications=password -o PubkeyAuthentication=no arij@192.168.159.144
arij@192.168.159.144: Permission denied (publickey).
arij@server26:~$
```

5- Installez Jenkins en tant que service et vérifiez son bon fonctionnement à partir de la machine physique :

a. Installer Java :

```
arij@server26:~$ sudo apt install -y fontconfig openjdk-21-jre
Installation de :
  fontconfig  openjdk-21-jre

Installation de dépendances :
  adwaita-icon-theme          libatk-wrapper-java      libglx-mesa0           librsvg2-2          libxfixes3
  alsa-topology-conf          libatk-wrapper-java-jni  libglx0                librsvg2-common     libxft2
  alsa-ucm-conf               libatk1.0-0t64           libglycin-2-0          libsharpyuv0        libxi6
  at-spi2-common              libatspi2.0-0t64         libgraphite2-3         libsm6              libxinerama1
  at-spi2-core                libavahi-client3         libgtk-3-0t64          libthai-data        libxkbfile1
  bubblewrap                  libavahi-common-data     libgtk-3-bin           libthai0            libxmu6
  ca-certificates-java        libavahi-common3         libgtk-3-common        libvulkan1          libxpm4
  dconf-gsettings-backend     libcairo-gobject2        libharfbuzz0b          libwayland-client0  libxrandr2
  dconf-service               libcairo2                libheif-plugin-aomdec  libwayland-cursor0  libxrender1
  fontconfig-config           libcolord2               libheif-plugin-aomenc  libwayland-egl1     libxres1
  fonts-dejavu-core           libcups2t64              libheif1               libx11-xcb1         libxshmfence1
  fonts-dejavu-extra          libdatrie1               libhwylt64             libxaw7             libxt6t64
  fonts-dejavu-mono           libdconf1                libibus-1.0-5          libxcb-dri3-0       libxtst6
  glycin-loaders              libdisplay-info3         libice6                libxcb-glx0         libxv1
  glycin-thumbnailers         libdrm-intel1            libjpeg-turbo8         libxcb-present0     libxxf86dga1
  gsettings-desktop-schemas   libepoxy0                libjpeg8               libxcb-randr0       libxxf86vm1
  gtk-update-icon-cache       libfontconfig1           libjxl0.11             libxcb-render0      luit
  hicolor-icon-theme          libgbm1                  liblcms2-2             libxcb-shape0       mesa-libgallium
  ibus-gtk3                   libgdk-pixbuf-2.0-0      libpango-1.0-0         libxcb-shm0         mesa-vulkan-drivers
  java-common                 libgdk-pixbuf2.0-common  libpangocairo-1.0-0    libxcb-sync1        openjdk-21-jre-headless
  libaom3                     libgif7                  libpangoft2-1.0-0      libxcb-xfixes0      user-session-migration
  libasound2-data             libgl1                   libpciaccess0          libxcomposite1      x11-common
  libasound2t64               libgl1-mesa-dri          libpcsclite1           libxcursor1         x11-utils
  libatk-bridge2.0-0t64       libglvnd0                libpixman-1-0          libxdamage1
```

```
arij@server26:~$ java -version
openjdk version "21.0.12.1" 2026-08-18
OpenJDK Runtime Environment (build 21.0.12.1+1-1-26.04.4-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.12.1+1-1-26.04.4-Ubuntu, mixed mode, sharing)
arij@server26:~$
```

B .Ajouter le dépôt Jenkins + installer

Clé GPG Jenkins

```
arij@server26:~$ sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
  https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
--2026-09-30 09:23:00--  https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
Resolving pkg.jenkins.io (pkg.jenkins.io)... 199.232.142.133, 2a04:4e42:63::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|199.232.142.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1680 (1,6K) [application/octet-stream]
Saving to: '/etc/apt/keyrings/jenkins-keyring.asc'

/etc/apt/keyrings/jenkins-keyring 100%[=====================================================>]   1,64K  --.-KB/s    in 0s

2026-09-30 09:23:01 (3,87 MB/s) - '/etc/apt/keyrings/jenkins-keyring.asc' saved [1680/1680]
```

c.Dépôt Jenkins

```
arij@server26:~$ echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc]" \
  https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
  /etc/apt/sources.list.d/jenkins.list > /dev/null
```

d.Installation

```
arij@server26:~$ sudo apt update
Atteint : 1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Atteint : 2 http://tn.archive.ubuntu.com/ubuntu resolute InRelease
Atteint : 3 http://tn.archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign : 4 https://pkg.jenkins.io/debian-stable binary/ InRelease
Atteint : 5 http://tn.archive.ubuntu.com/ubuntu resolute-backports InRelease
Atteint : 6 https://download.docker.com/linux/ubuntu resolute InRelease
Réception de : 7 https://pkg.jenkins.io/debian-stable binary/ Release [2 044 B]
Réception de : 8 https://pkg.jenkins.io/debian-stable binary/ Release.gpg [833 B]
Réception de : 9 https://pkg.jenkins.io/debian-stable binary/ Packages [31,8 kB]
34,7 ko réceptionnés en 2s (21,6 ko/s)
Tous les paquets sont à jour.
```

```
arij@server26:~$ sudo apt install -y jenkins
Installation de :
  jenkins

Installation de dépendances :
  net-tools

Sommaire :
  Mise à niveau de : 0. Installation de : 2, Supprimé : 0. Non mis à jour : 0
Taille du téléchargement : 53,9 Mo
  Espace nécessaire : 54,7 Mo / 14,9 Go disponible

Réception de : 1 http://tn.archive.ubuntu.com/ubuntu resolute/main amd64 net-tools amd64 2.10-2ubuntu1 [151 kB]
Réception de : 2 https://pkg.jenkins.io/debian-stable binary/ jenkins 2.580.1 [53,7 MB]
53,9 Mo réceptionnés en 1min 27s (620 ko/s)
Sélection du paquet net-tools précédemment désélectionné.
(Lecture de la base de données... 144139 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de .../net-tools_2.10-2ubuntu1_amd64.deb ...
Dépaquetage de net-tools (2.10-2ubuntu1) ...
Sélection du paquet jenkins précédemment désélectionné.
Préparation du dépaquetage de .../jenkins_2.580.1_all.deb ...
Dépaquetage de jenkins (2.580.1) ...
Paramétrage de net-tools (2.10-2ubuntu1) ...
Paramétrage de jenkins (2.580.1) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/jenkins.service' → '/usr/lib/systemd/system/jenkins.service'.
Traitement des actions différées (« triggers ») pour man-db (2.13.1-1build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.
```

e.Démarrer + activer le service

```
arij@server26:~$ sudo systemctl start jenkins
arij@server26:~$ sudo systemctl enable jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
arij@server26:~$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-09-30 09:27:23 UTC; 1min 40s ago
 Invocation: bcee21822e1b4b98a7fafd840ef22360
   Main PID: 4987 (java)
      Tasks: 43 (limit: 3937)
     Memory: 552.2M (peak: 567.5M)
        CPU: 36.414s
     CGroup: /system.slice/jenkins.service
             └─4987 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpP>

sept. 30 09:26:23 server26 jenkins[4987]: [LF]> This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
sept. 30 09:26:23 server26 jenkins[4987]: [LF]>
sept. 30 09:26:23 server26 jenkins[4987]: [LF]> ************************************************************
sept. 30 09:26:23 server26 jenkins[4987]: [LF]> ************************************************************
sept. 30 09:26:23 server26 jenkins[4987]: [LF]> ************************************************************
sept. 30 09:27:23 server26 jenkins[4987]: 2026-09-30 09:27:23.446+0000 [id=40]        INFO        jenkins.InitReactorRunner$1#onAtta>
sept. 30 09:27:23 server26 jenkins[4987]: 2026-09-30 09:27:23.480+0000 [id=30]        INFO        hudson.lifecycle.Lifecycle#onReady>
sept. 30 09:27:23 server26 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
sept. 30 09:27:25 server26 jenkins[4987]: 2026-09-30 09:27:25.482+0000 [id=56]        INFO        h.m.DownloadService$Downloadable#l>
sept. 30 09:27:25 server26 jenkins[4987]: 2026-09-30 09:27:25.488+0000 [id=56]        INFO        hudson.util.Retrier#start: Perform>
arij@server26:~$
```

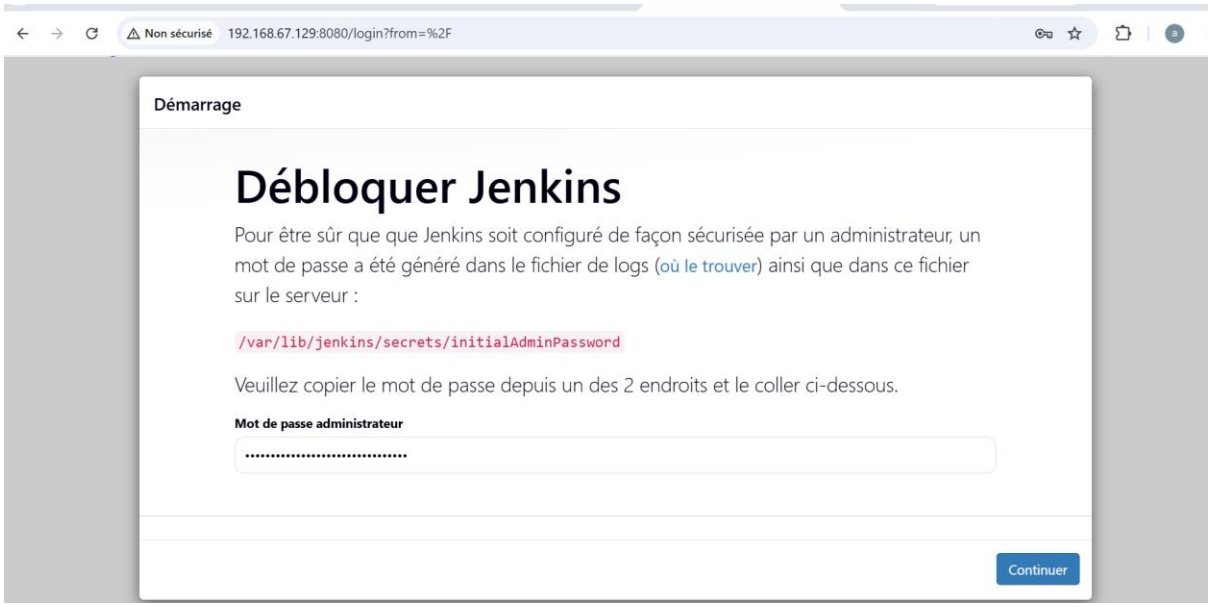
f. **Ouvrir le port 8080 dans le pare-feu**

```
arij@server26:~$ sudo ufw allow 8080/tcp
Rules updated
Rules updated (v6)
```

g. **Récupérer le mot de passe initial**

```
arij@server26:~$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
993e59e624be4ba1bacea6ff67ee9917
arij@server26:~$
```

h. **Test depuis Windows**


Non sécurisé
192.168.67.129:8080/login?from=%2F
Démarrage
Débloquer Jenkins
Pour être sûr que que Jenkins soit configuré de façon sécurisée par un administrateur, un mot de passe a été généré dans le fichier de logs (où le trouver) ainsi que dans ce fichier sur le serveur :
/var/lib/jenkins/secrets/initialAdminPassword
Veuillez copier le mot de passe depuis un des 2 endroits et le coller ci-dessous.
Mot de passe administrateur
Continuer


Démarrage
Configuration de l'instance
URL de Jenkins :
http://192.168.67.129:8080/
L'URL de Jenkins est utilisée pour fournir l'URL de base pour les liens absolus vers les diverses ressources Jenkins. Cela signifie que cette valeur est nécessaire pour le bon fonctionnement de nombreuses fonctionnalités de Jenkins, notamment les notifications par mail, les mises à jour des statuts des pull requests, et la variable d'environnement BUILD_URL fournie pour les étapes de build.
La valeur par défaut affichée n'est pas encore sauvegardée et est générée à partir de la requête actuelle, lorsque c'est possible. Il est fortement recommandé d'utiliser comme valeur l'URL qui est censée être utilisée par les utilisateurs. Cela évitera des confusions lors du partage ou de la visualisation de liens.
Jenkins 2.580.1
Passer cette étape et terminer
Sauver et terminer


Non sécurisé
192.168.67.129:8080
Jenkins
Nouveau Item
Historique des constructions
File d'attente des constructions
File d'attente des constructions vide
État du lanceur de compilations
0/2
Ajouter une description
Bienvenue sur Jenkins !
Vos jobs Jenkins seront affichés sur cette page. Pour commencer, vous pouvez mettre en place un build distribué ou commencer à créer un projet.
Commencer à créer votre projet
Créer un job
Configurer un build distribué
Mettre en place un agent
Configurer un cloud
En apprendre plus sur les builds distribués

6. Créez un mini CV One Page (HTML5/CSS3/Javascript) dans un dossier géré avec git et fournir son lien GitHub.

a. Créer un dossier pour le projet :

```
arij@server26:~$ mkdir -p ~/cv-onepage
arij@server26:~$ cd ~/cv-onepage
arij@server26:~/cv-onepage$
```

B.vérifier git est installé :

```
arij@server26:~/cv-onepage$ git --version
git version 2.53.0
```

C.Configurer Git :

```
arij@server26:~/cv-onepage$ git config --global user.name "arijbelaid"
arij@server26:~/cv-onepage$ git config --global user.email "aarijbelaid@gmail.com"
arij@server26:~/cv-onepage$ git config --list
user.name=arijbelaid
user.email=aarijbelaid@gmail.com
arij@server26:~/cv-onepage$
```

d.Créer les fichiers du CV :

```
arij@server26:~/cv-onepage$ nano index.html
arij@server26:~/cv-onepage$ nano style.css
arij@server26:~/cv-onepage$ nano script.js
```

Index.html

```
 GNU nano 8.7.1                                  index.html *
<!DOCTYPE html>
<html lang="fr">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>CV - Arij</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <header>
        <h1>Arij [Nom]</h1>
        <p class="subtitle">Étudiant(e) en Informatique</p>
    </header>

    <main>
        <section id="about">
            <h2>À propos</h2>
            <p>Passionné(e) par le DevOps, les systèmes Linux et l'automatisation.</p>
        </section>

        <section id="skills">
            <h2>Compétences</h2>
            <ul>
                <li>Linux / Ubuntu Server</li>
                <li>Docker</li>
                <li>Jenkins (CI/CD)</li>
                <li>Git / GitHub</li>
                <li>HTML5 / CSS3 / JavaScript</li>
            </ul>

^G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T Execute   ^C Location  M-U Undo      M-A Set Mark
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo      M-6 Copy
```

Style.css

```
 GNU nano 8.7.1                                   style.css
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: 'Segoe UI', Arial, sans-serif;
    line-height: 1.6;
    color: #333;
    background: #f4f4f4;
    padding: 20px;
}

header {
    background: linear-gradient(135deg, #667eea, #764ba2);
    color: white;
    padding: 40px 20px;
    border-radius: 10px;
    text-align: center;
    margin-bottom: 20px;
}

header h1 {
    font-size: 2.5em;
    margin-bottom: 10px;
}

.subtitle {
```

Script.js

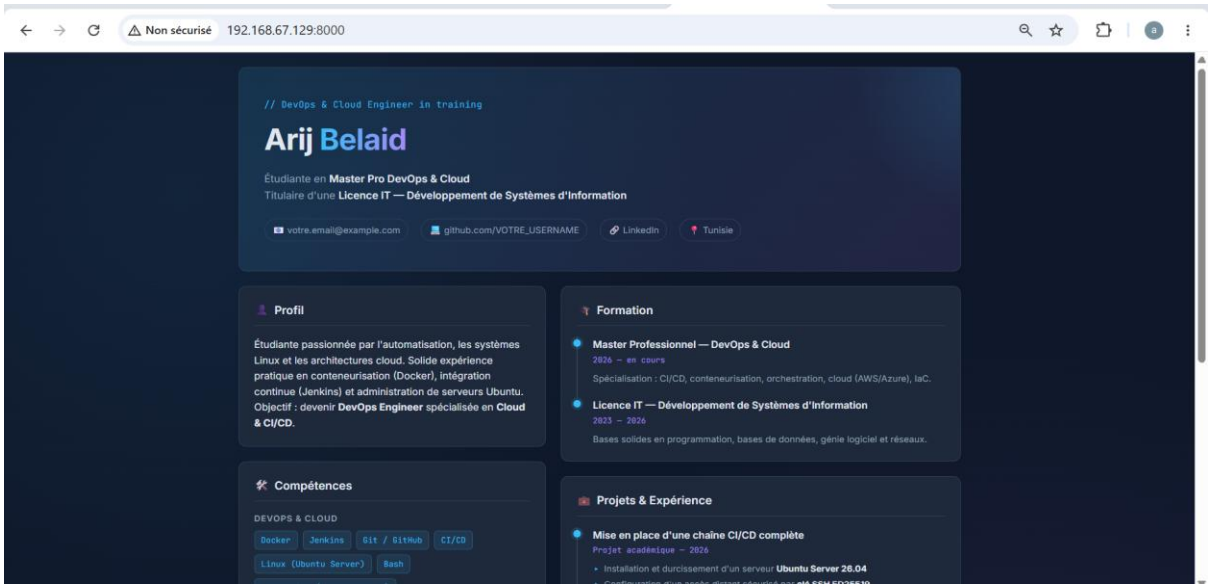
```
 GNU nano 8.7.1                                          script.js
// Animation d'apparition des sections au scroll
document.addEventListener('DOMContentLoaded', () => {
    const sections = document.querySelectorAll('section');

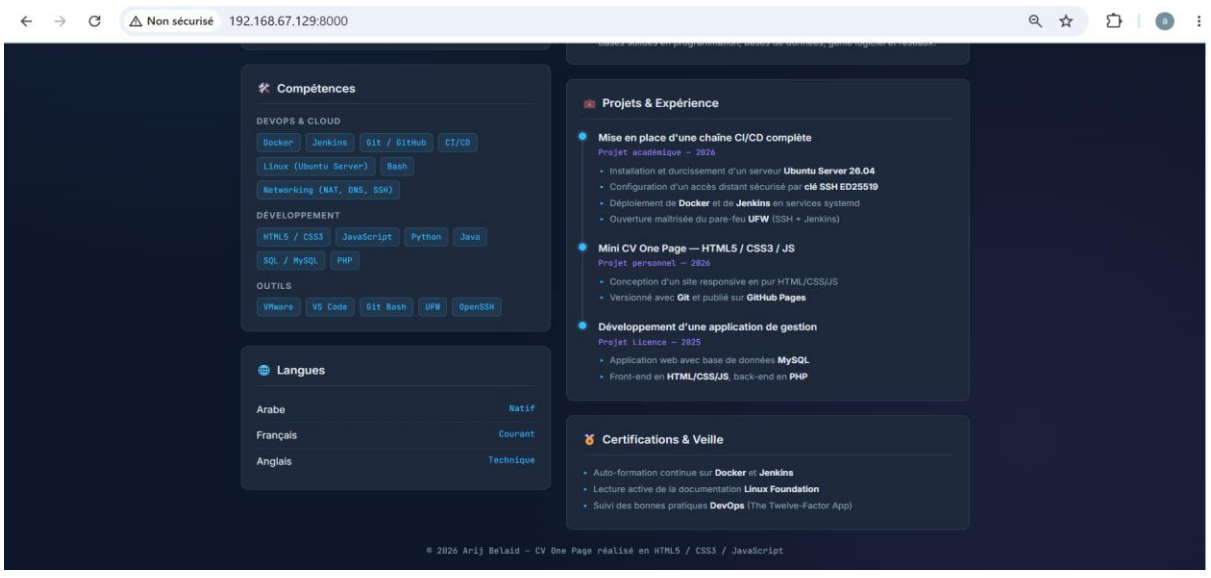
    const observer = new IntersectionObserver((entries) => {
        entries.forEach(entry => {
            if (entry.isIntersecting) {
                entry.target.style.opacity = '1';
                entry.target.style.transform = 'translateY(0)';
            }
        });
    }, { threshold: 0.1 });

    sections.forEach(section => {
        section.style.opacity = '0';
        section.style.transform = 'translateY(20px)';
        section.style.transition = 'opacity 0.6s ease, transform 0.6s ease';
        observer.observe(section);
    });

    // Affichage de l'année dynamique
    const year = new Date().getFullYear();
    document.querySelector('footer p').innerHTML = `&copy; ${year} Arij. Tous droits réservés.`;
});
```

Voici cv





6- Activer les push GitHub via SSH : Ouvrez GitHub / Settings → SSH and GPG keys / Cliquez sur New SSH key / Donnez un nom à la clé / Collez votre clé publique / Enregistrez / Tester: ssh -T git@github.com, Configurer le dépôt local pour utiliser SSH.

a.Générer et ajouter la clé SSH sur votre VM : Vérifier les clés existantes

```
arij@server26:~/cv-onepage$ ls -al ~/.ssh
total 20
drwx------ 2 arij arij 4096 Sep 29 21:29 .
drwxr-x--- 6 arij arij 4096 Sep 30 10:00 ..
-rw------- 1 arij arij  192 Sep 29 21:31 authorized_keys
-rw------- 1 arij arij  978 Sep 29 21:29 known_hosts
-rw-r--r-- 1 arij arij  142 Sep 29 21:29 known_hosts.old
arij@server26:~/cv-onepage$
```

b.Générer une nouvelle paire de clés SSH :

```
arij@server26:~/cv-onepage$ ssh-keygen -t ed25519 -C "aarijbelaid@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/arij/.ssh/id_ed25519):
Enter passphrase for "/home/arij/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/arij/.ssh/id_ed25519
Your public key has been saved in /home/arij/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:nmhBXDM62NnfrRGxr0q+hx2LnSvJ/siUE7hsJ9RGkQo aarijbelaid@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|        + ...    |
|     +E= o.. o   |
|    . B.... o    |
|     . ..= . +   |
|      . S = o o  |
|       * + o.+   |
|      o B.**o+   |
|     . . B***    |
|         .B=o.   |
+----[SHA256]-----+
arij@server26:~/cv-onepage$
```

c.Afficher la clé publique :

```
arij@server26:~/cv-onepage$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIIfeNBGqKovV1Z1ttYTiE18Jf2jPGyzG8c+cJ2qhPJml aarijbelaid@gmail.com
arij@server26:~/cv-onepage$
```

d.Tester la connexion SSH :

```
arij@server26:~/cv-onepage$ ssh -T git@github.com
The authenticity of host 'github.com (140.82.121.3)' can't be established.
ED25519 key fingerprint is: SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Hi arijbelaid! You've successfully authenticated, but GitHub does not provide shell access.
arij@server26:~/cv-onepage$
```

e.Configurer le dépôt local pour utiliser SSH

```
arij@server26:~/cv-onepage$ git remote set-url origin git@github.com:arijbelaid/cv-onepage.git
arij@server26:~/cv-onepage$ git remote -v
origin  git@github.com:arijbelaid/cv-onepage.git (fetch)
origin  git@github.com:arijbelaid/cv-onepage.git (push)
arij@server26:~/cv-onepage$ nano ~/cv-onepage/README.md
arij@server26:~/cv-onepage$ git add .
arij@server26:~/cv-onepage$ git commit -m "docs: ajout README avec commandes SSH et capture"
[main a05a1e6] docs: ajout README avec commandes SSH et capture
 1 file changed, 76 insertions(+)
 create mode 100644 README.md
arij@server26:~/cv-onepage$ git push -u origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 2 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (8/8), 6.31 KiB | 538.00 KiB/s, done.
Total 8 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To github.com:arijbelaid/cv-onepage.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
arij@server26:~/cv-onepage$
```

Évolution du mini-CV vers DevSecOps Portfolio

7- Faites évoluer le mini CV en une petite application DevSecOps Portfolio comportant au minimum les sections : About / Skills / Projects / Experience / Contact.
README.md : ajoutez une capture de la nouvelle version et décrivez brièvement les principales améliorations.

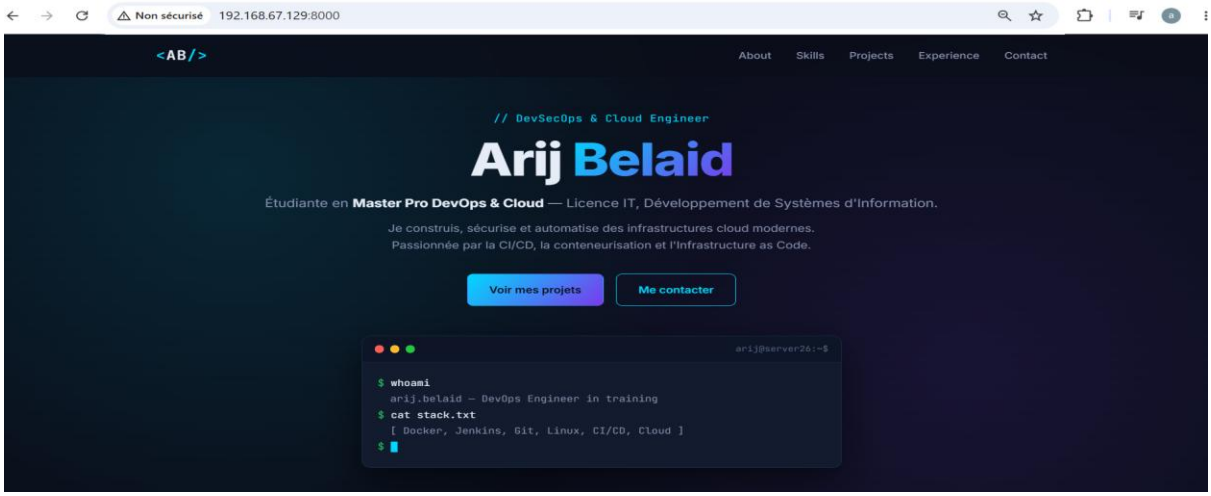
a. Réécrire index.html ,style.css et script.js en version DevSecOps Portfolio

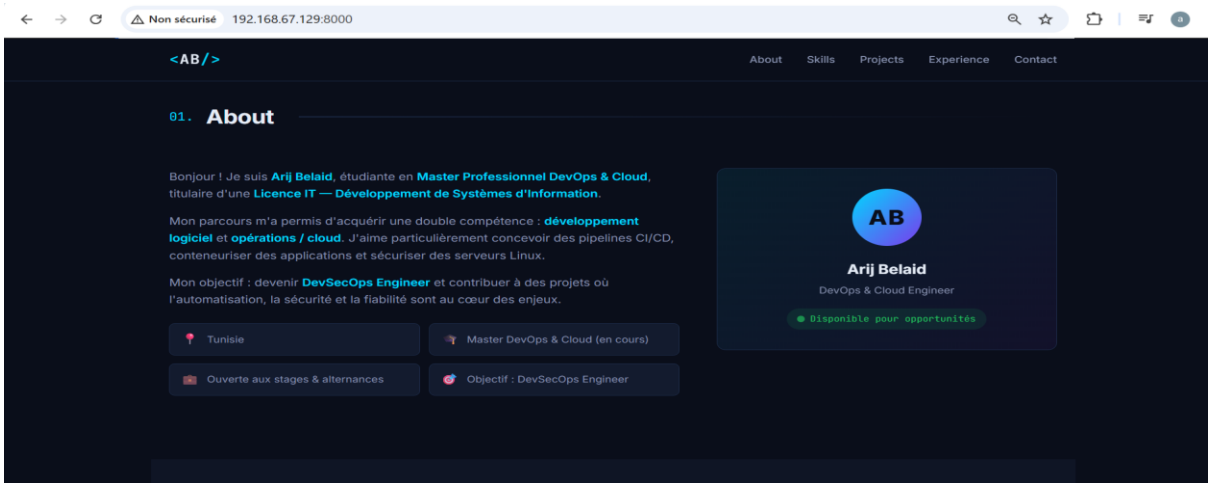
```
arij@server26:~/cv-onepage$ nano index.html
arij@server26:~/cv-onepage$ nano style.css
arij@server26:~/cv-onepage$ nano script.js
arij@server26:~/cv-onepage$ nano README.md
```

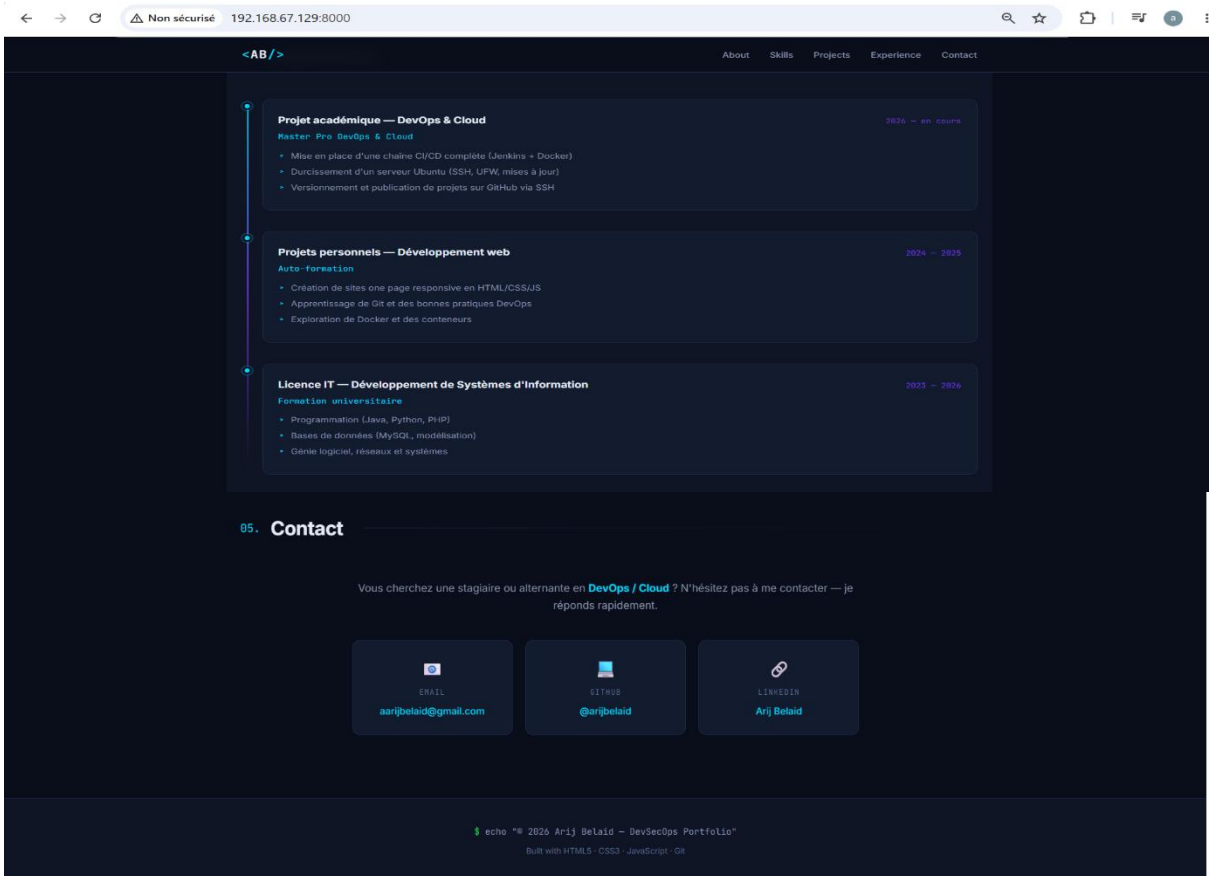
b. Lancer en local

```
arij@server26:~/cv-onepage$ git clone git@github.com:arijbelaid/cv-onepage.git
Cloning into 'cv-onepage'...
remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (7/7), done.
remote: Total 8 (delta 1), reused 8 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (8/8), 6.31 KiB | 3.16 MiB/s, done.
Resolving deltas: 100% (1/1), done.
arij@server26:~/cv-onepage$ python3 -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
192.168.67.1 - - [03/Oct/2026 21:12:55] "GET / HTTP/1.1" 200 -
192.168.67.1 - - [03/Oct/2026 21:12:55] "GET /style.css HTTP/1.1" 200 -
```

Cv :







c.Transférer la capture dans la VM :

```
C:\Users\arijb>scp %USERPROFILE%\Desktop\screenshot.png arij@192.168.67.129:~/cv-onepage/screenshot.png
screenshot.png                                                100%  474KB  46.3MB/s   00:00
```

```
arij@server26:~/cv-onepage$ ls -la ~/cv-onepage/screenshot.png
-rw-rw-r-- 1 arij arij 485435 Oct  3 21:32 /home/arij/cv-onepage/screenshot.png
arij@server26:~/cv-onepage$ file ~/cv-onepage/screenshot.png
/home/arij/cv-onepage/screenshot.png: PNG image data, 1884 x 913, 8-bit/color RGBA, non-interlaced
arij@server26:~/cv-onepage$
```

d. Commit + Push

```
arij@server26:~/cv-onepage$ git push
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 2 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 469.99 KiB | 3.38 MiB/s, done.
Total 7 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:arijbelaid/cv-onepage.git
   a05a1e6..f57ce9e  main -> main
arij@server26:~/cv-onepage$
```

e/ajout de Démo en ligne

https://arijbelaid.github.io/cv-onepage/

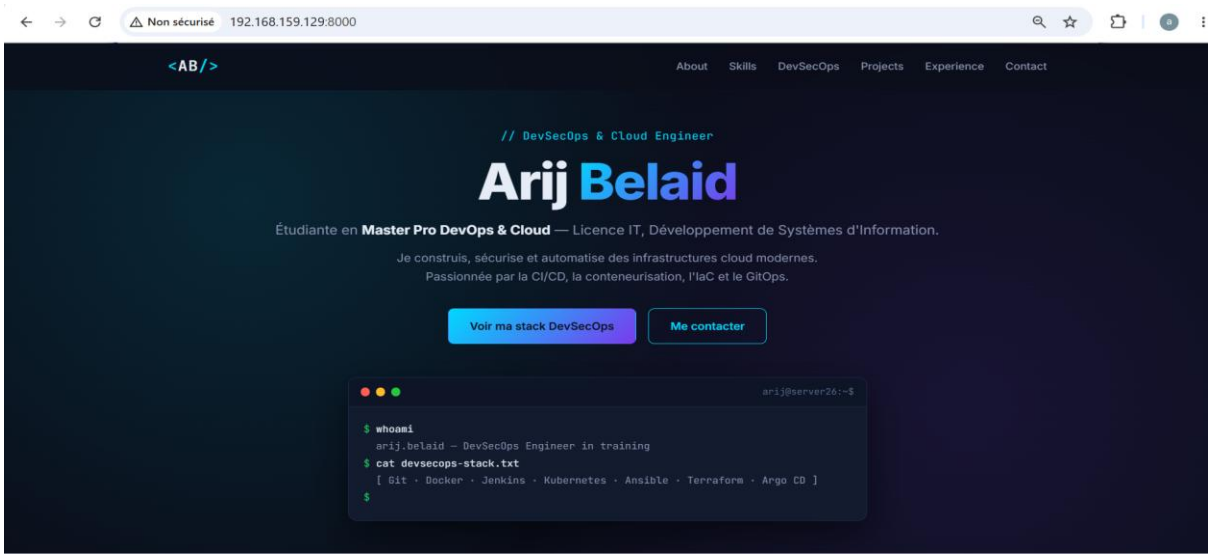
8- Ajoutez une section DevSecOps Skills permettant d'afficher les technologies utilisées pendant le projet : Git / Docker / Jenkins / Kubernetes / Ansible / Terraform / Argo CD

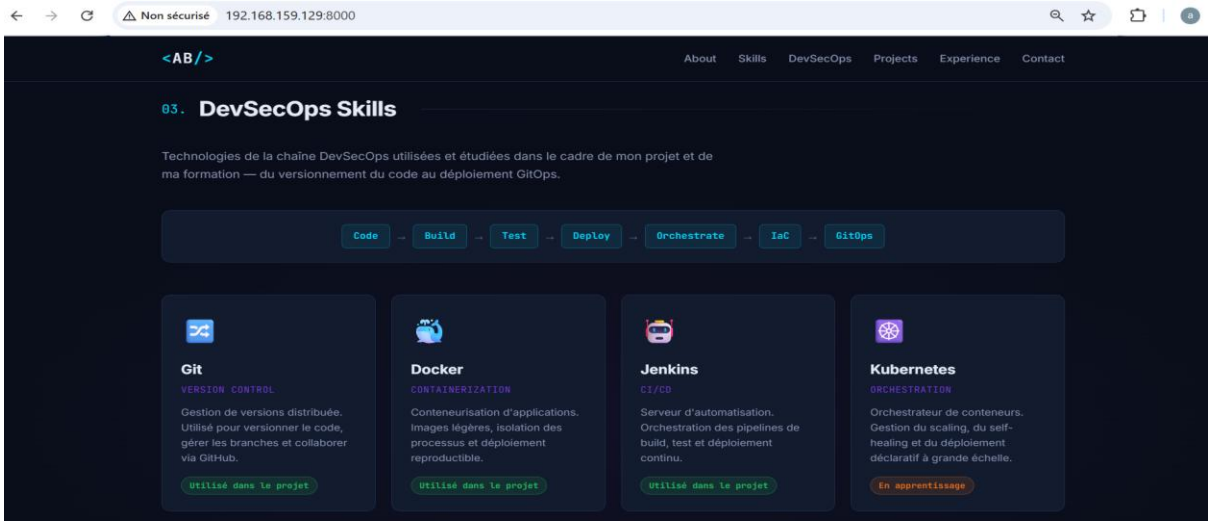
README.md : ajoutez une capture d'écran.

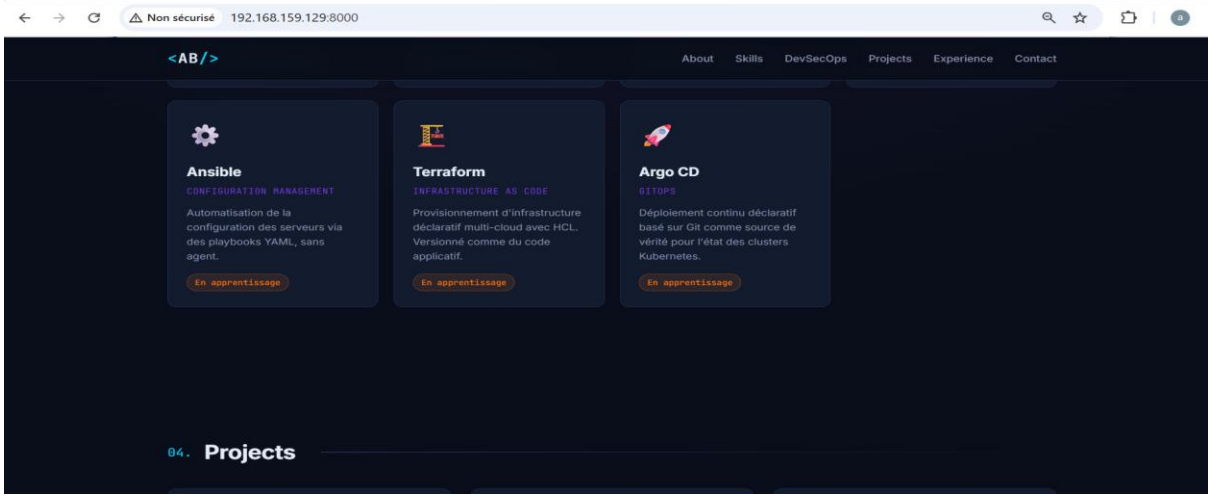
a. Réécrire index.html ,style.css et script.js :

```
arij@server26:~/cv-onepage$ nano index.html
arij@server26:~/cv-onepage$ nano style.css
arij@server26:~/cv-onepage$ nano script.js
```

b.cv partie ajoutée DevSecOps :







***Transférer la capture de partie DevSecOps vers la VM :**

```
C:\Users\arijb>scp %USERPROFILE%\Desktop\screenshot1.png arij@192.168.159.129:~/cv-onepage/screenshot1.png
screenshot1.png                                                                    100%  229KB  18.7MB/s   00:00

C:\Users\arijb>
```

```
arij@server26:~/cv-onepage$ ls -la ~/cv-onepage/screenshot1.png
-rw-rw-r-- 1 arij arij 234824 Oct  6 08:51 /home/arij/cv-onepage/screenshot1.png
```

c. Commit + Push :

```
arij@server26:~/cv-onepage$ git add .
arij@server26:~/cv-onepage$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md
        modified:   index.html
        new file:   screenshot1.png
        modified:   script.js
        modified:   style.css

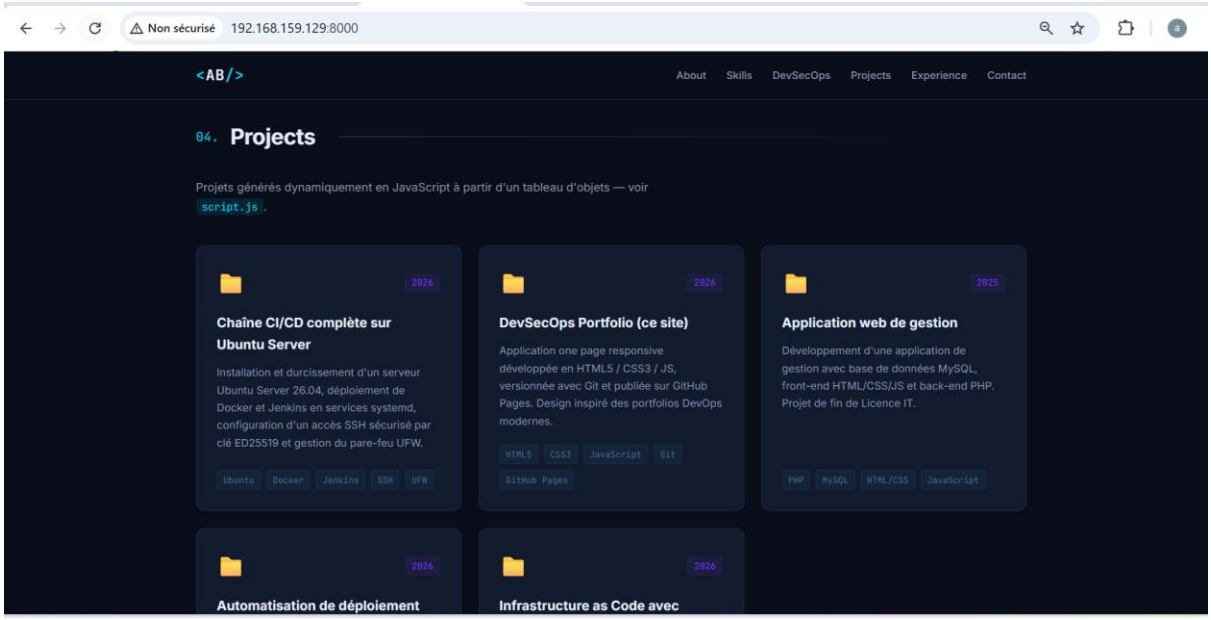
arij@server26:~/cv-onepage$ git commit -m "feat(v3): ajout section DevSecOps Skills (Git/Docker/Jenkins/K8s/Ansible/Terraform/Argo CD
) + capture"
[main 1a281bf] feat(v3): ajout section DevSecOps Skills (Git/Docker/Jenkins/K8s/Ansible/Terraform/Argo CD) + capture
 5 files changed, 364 insertions(+), 76 deletions(-)
 create mode 100644 screenshot1.png
arij@server26:~/cv-onepage$ git push
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 2 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 222.21 KiB | 1.90 MiB/s, done.
Total 7 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To github.com:arijbelaid/cv-onepage.git
   f57ce9e..1a281bf  main -> main
arij@server26:~/cv-onepage$
```

9- Ajoutez une section Projects générée dynamiquement en JavaScript à partir d'un tableau d'objets.
README.md : ajoutez un extrait du code JavaScript et une capture du résultat.

a. Réécrire index.html ,style.css et script.js :

```
arij@server26:~/cv-onepage$ nano index.html
arij@server26:~/cv-onepage$ nano script.js
arij@server26:~/cv-onepage$ nano style.css
```

b. Tester en local :



c. Commit + Push :

```
arij@server26:~/cv-onepage$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIOS5flwXHyYq09On4HRPoHUCfX1No//Xyj74DHpxyiIJ aarijbelaid@gmail.com
arij@server26:~/cv-onepage$ ssh -T git@github.com
Hi arijbelaid! You've successfully authenticated, but GitHub does not provide shell access.
arij@server26:~/cv-onepage$ git push
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 4.74 KiB | 606.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:arijbelaid/cv-onepage.git
   1a281bf..10d1e68  main -> main
arij@server26:~/cv-onepage$
```

Dockerisation initiale :

10- Créez un Dockerfile pour servir votre DevSecOps Portfolio avec Nginx. README.md : ajoutez le contenu du Dockerfile et une courte explication.

a. Créer Dockerfile,ngnix.conf et Dockerignore

```
arij@server26:~/cv-onepage$ nano Dockerfile
arij@server26:~/cv-onepage$ nano nginx.conf
arij@server26:~/cv-onepage$ nano .dockerignore
```

b.Construire l'image Docker :

```
arij@server26:~/cv-onepage$ docker build -t cv-onepage:1.0 .
[+] Building 19.8s (12/12) FINISHED                                                
```

f. Tester l'accès depuis la VM :

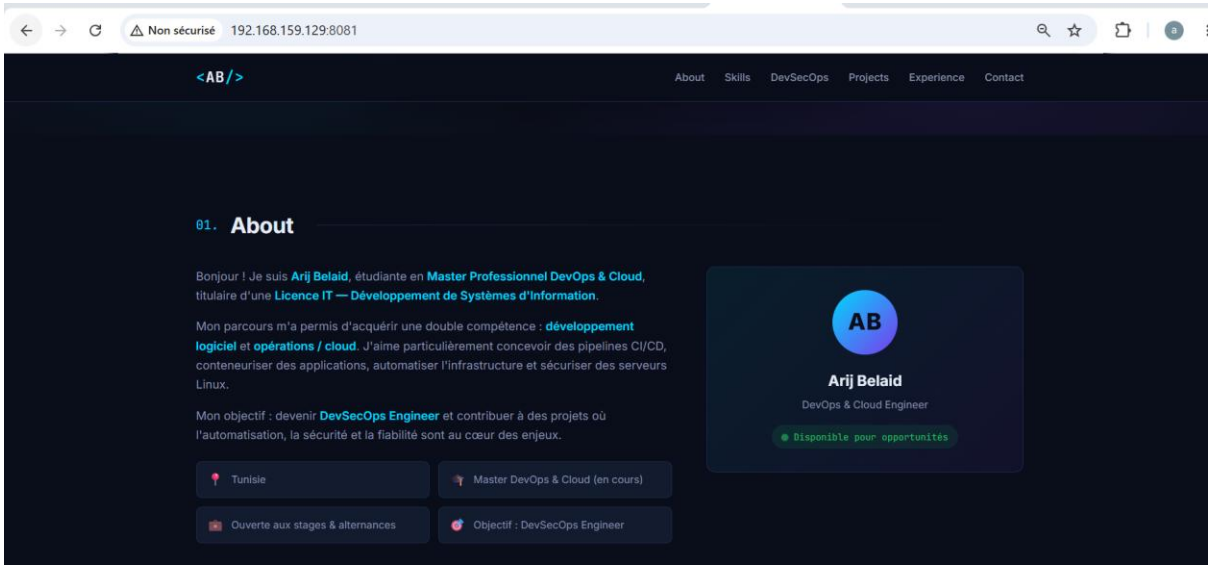
```
arij@server26:~/cv-onepage$ curl -I http://localhost:8081
HTTP/1.1 200 OK
Server: nginx
Date: Tue, 06 Oct 2026 09:59:41 GMT
Content-Type: text/html
Content-Length: 20201
Last-Modified: Tue, 06 Oct 2026 09:09:18 GMT
Connection: keep-alive
ETag: "6ac4babe-4ee9"
X-Frame-Options: SAMEORIGIN
X-Content-Type-Options: nosniff
X-XSS-Protection: 1; mode=block
Referrer-Policy: no-referrer-when-downgrade
Accept-Ranges: bytes
```

g. ouvrir le port 8081 dans UFW :

```
arij@server26:~/cv-onepage$ sudo ufw allow 8081/tcp
[sudo: authenticate] Password:
Rule added
Rule added (v6)
arij@server26:~/cv-onepage$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
8080/tcp                   ALLOW       Anywhere
22/tcp                     ALLOW       Anywhere
8000/tcp                   ALLOW       Anywhere
8081/tcp                   ALLOW       Anywhere
8080/tcp (v6)              ALLOW       Anywhere (v6)
22/tcp (v6)                ALLOW       Anywhere (v6)
8000/tcp (v6)              ALLOW       Anywhere (v6)
8081/tcp (v6)              ALLOW       Anywhere (v6)
```

h. Tester depuis Windows



i. Commit + push

```
arij@server26:~/cv-onepage$ git add .
arij@server26:~/cv-onepage$ git commit -m "feat(v5): Dockerfile Nginx + capture du portfolio conteneurisé"
[main 2e21e37] feat(v5): Dockerfile Nginx + capture du portfolio conteneurisé
 3 files changed, 72 insertions(+)
 create mode 100644 .dockerignore
 create mode 100644 Dockerfile
 create mode 100644 nginx.conf
arij@server26:~/cv-onepage$ git push
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 1.48 KiB | 379.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:arijbelaid/cv-onepage.git
   10d1e68..2e21e37  main -> main
```

11- Construisez une image Docker nommée : cv-docker
README.md : ajoutez la commande utilisée et une capture d'écran du résultat.

a. Construire l'image cv-docker :

```
arij@server26:~/cv-onepage$ docker build -t cv-docker .
[+] Building 1.9s (12/12) FINISHED                                                                                  docker:default
 => [internal] load build definition from Dockerfile                                                                          0.0s
 => => transferring dockerfile: 1.09kB                                                                                        0.0s
 => [internal] load metadata for docker.io/library/nginx:alpine                                                               1.5s
 => [internal] load .dockerignore                                                                                             0.0s
 => => transferring context: 128B                                                                                             0.0s
 => [1/7] FROM docker.io/library/nginx:alpine@sha256:df221db836e1754089190208cee7eeda94f233197056426eda74a43ab1abeac2         0.0s
 => => resolve docker.io/library/nginx:alpine@sha256:df221db836e1754089190208cee7eeda94f233197056426eda74a43ab1abeac2         0.0s
 => [internal] load build context                                                                                             0.0s
 => => transferring context: 157B                                                                                             0.0s
 => CACHED [2/7] RUN rm -rf /usr/share/nginx/html/*                                                                           0.0s
 => CACHED [3/7] COPY index.html    /usr/share/nginx/html/                                                                    0.0s
 => CACHED [4/7] COPY style.css     /usr/share/nginx/html/                                                                    0.0s
 => CACHED [5/7] COPY script.js     /usr/share/nginx/html/                                                                    0.0s
 => CACHED [6/7] COPY screenshot.png /usr/share/nginx/html/                                                                   0.0s
 => CACHED [7/7] COPY nginx.conf /etc/nginx/conf.d/default.conf                                                               0.0s
 => exporting to image                                                                                                        0.1s
 => => exporting layers                                                                                                       0.0s
 => => exporting manifest sha256:2ac6b25f9fe460d03e4e0a6c849d30ffbe767be8cacd2cbe0521c93859bccd48                             0.0s
 => => exporting config sha256:fb604bff9367746de84a4f705d8bf456eadaa5f42365ecca80018005b7aa0d637                              0.0s
 => => exporting attestation manifest sha256:9c2bbdd30856564a754459a94962e5a892ea08530a69034f0a5582cc0faa0ac0                 0.0s
 => => exporting manifest list sha256:fbb9c5eb486f3ee7d78fcc72486cbf02498feffeee3cb1b564feeeb2f6fd58e7                        0.0s
 => => naming to docker.io/library/cv-docker:latest                                                                           0.0s
 => => unpacking to docker.io/library/cv-docker:latest                                                                        0.0s
arij@server26:~/cv-onepage$
```

b. Vérifier que l'image existe

```
arij@server26:~/cv-onepage$ docker images | grep cv-docker
cv-docker:latest      fbb9c5eb486f       94.7MB         26.8MB    U
arij@server26:~/cv-onepage$
```

12- Exécutez un conteneur en exposant le Portfolio sur un port accessible et vérifiez l'accès depuis la machine physique.
README.md : ajoutez la commande docker run, le résultat de docker ps et une capture d'écran.

a. Lancer un conteneur cv-docker sur le port 8082

```
arij@server26:~/cv-onepage$ docker run -d --name cv-docker-container -p 8082:80 cv-docker
b902662c64aa4a446bb2a83c5f104727dbc712cf5f82c4a1366567350b5d4883
arij@server26:~/cv-onepage$
```

b. Vérifier que le conteneur tourne

```
arij@server26:~/cv-onepage$ docker ps
CONTAINER ID   IMAGE            COMMAND                  CREATED          STATUS          PORTS                                       N
AMES
b902662c64aa   cv-docker        "/docker-entrypoint.…"   41 seconds ago   Up 41 seconds   0.0.0.0:8082->80/tcp, [::]:8082->80/tcp     c
v-docker-container
3a8f6c145be1   cv-onepage:1.0   "/docker-entrypoint.…"   33 minutes ago   Up 33 minutes   0.0.0.0:8081->80/tcp, [::]:8081->80/tcp     c
v-portfolio
```

c. Ouvrir le port 8082 dans UFW

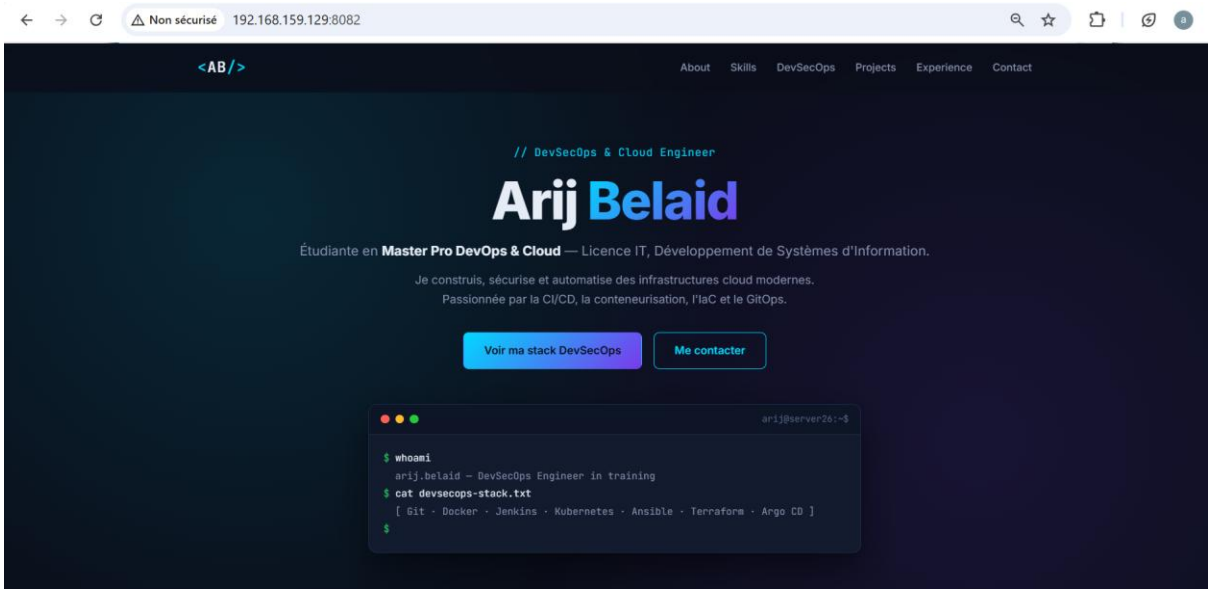
```
arij@server26:~/cv-onepage$ sudo ufw allow 8082/tcp
[sudo: authenticate] Password:
Rule added
Rule added (v6)
arij@server26:~/cv-onepage$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
8080/tcp                   ALLOW       Anywhere
22/tcp                     ALLOW       Anywhere
8000/tcp                   ALLOW       Anywhere
8081/tcp                   ALLOW       Anywhere
8082/tcp                   ALLOW       Anywhere
8080/tcp (v6)              ALLOW       Anywhere (v6)
22/tcp (v6)                ALLOW       Anywhere (v6)
8000/tcp (v6)              ALLOW       Anywhere (v6)
8081/tcp (v6)              ALLOW       Anywhere (v6)
8082/tcp (v6)              ALLOW       Anywhere (v6)
```

d. Tester en local (dans la VM)

```
arij@server26:~/cv-onepage$ curl -I http://localhost:8082
HTTP/1.1 200 OK
Server: nginx
Date: Tue, 06 Oct 2026 10:32:57 GMT
Content-Type: text/html
Content-Length: 20201
Last-Modified: Tue, 06 Oct 2026 09:09:18 GMT
Connection: keep-alive
ETag: "6ac4babe-4ee9"
X-Frame-Options: SAMEORIGIN
X-Content-Type-Options: nosniff
X-XSS-Protection: 1; mode=block
Referrer-Policy: no-referrer-when-downgrade
Accept-Ranges: bytes
```

e.Vérifier l'accès depuis la machine physique (Windows) :



f. Commit + Push :

```
arij@server26:~/cv-onepage$ git add .
arij@server26:~/cv-onepage$ git commit -m "feat(v7): exécution du conteneur cv-docker + accès depuis la machine physique"
[main ae7aa99] feat(v7): exécution du conteneur cv-docker + accès depuis la machine physique
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 screenshot2.png
arij@server26:~/cv-onepage$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 143.00 KiB | 1.18 MiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:arijbelaid/cv-onepage.git
   2e21e37..ae7aa99  main -> main
arij@server26:~/cv-onepage$
```

13- Déployez le Portfolio avec Docker Compose : *docker compose up -d* README.md : ajoutez la commande utilisée, le résultat de docker compose ps et une capture d'écran.

a.Créer le fichier docker-compose.yml :

```
arij@server26:~/cv-onepage$ nano docker-compose.yml
```

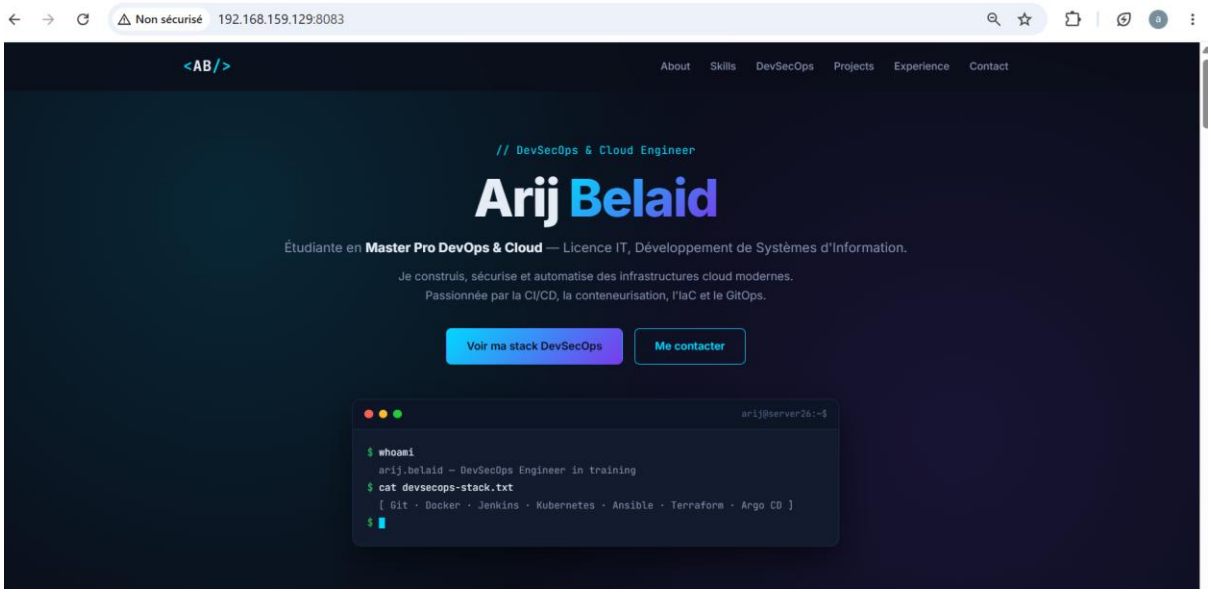
b. Lancer avec Docker Compose

```
arij@server26:~/cv-onepage$ docker compose up -d
[+] up 2/2
 ✔Network cv-onepage_default      Created                                    0.2s
 ✔Container cv-portfolio-compose  Started                                    1.0s
arij@server26:~/cv-onepage$
```

c. Vérifier avec docker compose ps

```
arij@server26:~/cv-onepage$ docker compose ps
NAME                 IMAGE              COMMAND                  SERVICE     CREATED          STATUS                    PORTS
cv-portfolio-compose   cv-docker:latest   "/docker-entrypoint.…"   portfolio   29 seconds ago   Up 28 seconds (healthy)   0.0.0.0:8083->80/tcp, [::]:8083->80/tcp
arij@server26:~/cv-onepage$
```

Test :



14- Publiez les modifications du projet sur GitHub via SSH.
README.md : ajoutez les commandes Git utilisées et le lien du dépôt GitHub mis à jour.

a. Vérifier l'état Git

```
arij@server26:~/cv-onepage$ git status
git log --oneline -10
git remote -v
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
3834006 (HEAD -> main, origin/main) feat(v8): déploiement avec Docker Compose + healthcheck healthy
ae7aa99 feat(v7): exécution du conteneur cv-docker + accès depuis la machine physique
2e21e37 feat(v5): Dockerfile Nginx + capture du portfolio conteneurisé
10d1e68 feat(v4): section Projects générée dynamiquement en JavaScript depuis un tableau d'objets
1a281bf feat(v3): ajout section DevSecOps Skills (Git/Docker/Jenkins/K8s/Ansible/Terraform/Argo CD) + capture
f57ce9e feat(v2): évolution DevSecOps Portfolio + capture d'écran
a05a1e6 docs: ajout README avec commandes SSH et capture
a5ebb1b feat: CV one page Arij Belaid - HTML5/CSS3/JS
origin  git@github.com:arijbelaid/cv-onepage.git (fetch)
origin  git@github.com:arijbelaid/cv-onepage.git (push)
arij@server26:~/cv-onepage$
```

```
arij@server26:~/cv-onepage$ nano README.md
arij@server26:~/cv-onepage$ git add README.md
arij@server26:~/cv-onepage$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md

arij@server26:~/cv-onepage$ git commit -m "docs: ajout section Git/SSH et lien dépôt (Question 14)"
[main 4df6e62] docs: ajout section Git/SSH et lien dépôt (Question 14)
 1 file changed, 76 insertions(+)
arij@server26:~/cv-onepage$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.38 KiB | 707.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:arijbelaid/cv-onepage.git
   3834006..4df6e62  main -> main
```

Première introduction à l'automatisation :

15- Installez Vagrant (dans la VM) et créez un premier Vagrantfile permettant de créer automatiquement une VM Ubuntu Server (vérifiez que votre VM dispose d'assez de RAM et d'espace disque).

a. *Mettre à jour*

```
arij@server26:~$ sudo apt update
[sudo: authenticate] Password:
Atteint : 1 http://tn.archive.ubuntu.com/ubuntu resolute InRelease
Réception de : 2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Ign : 3 https://pkg.jenkins.io/debian-stable binary/ InRelease
Réception de : 4 http://tn.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Atteint : 5 https://pkg.jenkins.io/debian-stable binary/ Release
Réception de : 7 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [597 kB]
Réception de : 8 http://tn.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Réception de : 9 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [137 kB]
Réception de : 10 http://tn.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [729 kB]
Réception de : 11 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46,7 kB]
Réception de : 12 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [474 kB]
Réception de : 13 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [96,3 kB]
Réception de : 14 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [199 kB]
Réception de : 15 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [64,8 kB]
Réception de : 16 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [55,5 kB]
Réception de : 17 https://download.docker.com/linux/ubuntu resolute InRelease [32,5 kB]
Réception de : 18 http://tn.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [171 kB]
Réception de : 19 http://tn.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97,7 kB]
Réception de : 20 http://tn.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [320 kB]
Réception de : 21 http://tn.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [104 kB]
Réception de : 22 http://tn.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [201 kB]
Réception de : 23 http://tn.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1 056 B]
3 739 ko réceptionnés en 7s (561 ko/s)
21 paquets peuvent être mis à jour. Exécutez « apt list --upgradable » pour les voir.
arij@server26:~$
```

b. *Clé GPG HashiCorp :*

```
arij@server26:~$ ls -la /usr/share/keyrings/hashicorp-archive-keyring.gpg
-rw-r--r-- 1 root root 1213 Oct  6 21:12 /usr/share/keyrings/hashicorp-archive-keyring.gpg
```

c. *ajout du dépôt HashiCorp +vérification :*

```
arij@server26:~$ cat /etc/apt/sources.list.d/hashicorp.list
deb [signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.releases.hashicorp.com resolute main
arij@server26:~$
```

d. *Mettre à jour*

```
arij@server26:~$ sudo apt update
Réception de : 1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Atteint : 2 http://tn.archive.ubuntu.com/ubuntu resolute InRelease
Atteint : 3 http://tn.archive.ubuntu.com/ubuntu resolute-updates InRelease
Atteint : 4 https://download.docker.com/linux/ubuntu resolute InRelease
Atteint : 5 http://tn.archive.ubuntu.com/ubuntu resolute-backports InRelease
Réception de : 6 https://apt.releases.hashicorp.com resolute InRelease [12,9 kB]
Ign : 7 https://pkg.jenkins.io/debian-stable binary/ InRelease
Réception de : 8 https://apt.releases.hashicorp.com resolute/main amd64 Packages [324 kB]
Réception de : 9 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [597 kB]
Atteint : 10 https://pkg.jenkins.io/debian-stable binary/ Release
Réception de : 12 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [137 kB]
1 209 ko réceptionnés en 2s (637 ko/s)
21 paquets peuvent être mis à jour. Exécutez « apt list --upgradable » pour les voir.
```

e. *installer Vagrant*

```
arij@server26:~$ sudo apt install -y vagrant
Installation de :
  vagrant

Sommaire :
  Mise à niveau de : 0. Installation de : 1, Supprimé : 0. Non mis à jour : 21
Taille du téléchargement : 90,6 Mo
  Espace nécessaire : 274 Mo / 14,1 Go disponible

Réception de : 1 https://apt.releases.hashicorp.com resolute/main amd64 vagrant amd64 2.4.9-1 [90,6 MB]
90,6 Mo réceptionnés en 17s (5 431 ko/s)
Sélection du paquet vagrant précédemment désélectionné.
(Lecture de la base de données... 148296 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de .../vagrant_2.4.9-1_amd64.deb ...
Dépaquetage de vagrant (2.4.9-1) ...
Paramétrage de vagrant (2.4.9-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

f. Vérifier l'installation de Vagrant

```
arij@server26:~$ vagrant --version
Vagrant 2.4.9
arij@server26:~$
```

g. Installer VirtualBox :

```
arij@server26:~$ sudo apt install -y virtualbox
Installation de :
  virtualbox

Installation de dépendances :
  build-essential            libqt6dbus6
  bzip2                      libqt6gui6
  cpp                        libqt6help6
  cpp-15                     libqt6network6
  cpp-15-x86-64-linux-gnu    libqt6opengl6
  cpp-x86-64-linux-gnu       libqt6printsupport6
  dkms                       libqt6qml6
  dpkg-dev                   libqt6qmlmeta6
  fakeroot                   libqt6qmlmodels6
  g++                        libqt6qmlworkerscript6
  g++-15                     libqt6quick6
  g++-15-x86-64-linux-gnu    libqt6sql6
  g++-x86-64-linux-gnu       libqt6sql6-sqlite
  gcc                        libqt6statemachine6
  gcc-15                     libqt6svg6
```

```
arij@server26:~$ VBoxManage --version
7.2.6_Ubuntur172322
arij@server26:~$
```

h.Créer le dossier + Vagrantfile :

```
arij@server26:~$ mkdir -p ~/vagrant-ubuntu
arij@server26:~$ cd ~/vagrant-ubuntu
arij@server26:~/vagrant-ubuntu$ nano Vagrantfile
```

```
Vagrant.configure("2") do |config|
  config.vm.box = "generic/ubuntu2204"
  config.vm.hostname = "vagrant-ubuntu"

  config.vm.provider "virtualbox" do |vb|
    vb.memory = 1024
    vb.cpus   = 1
    vb.customize ["modifyvm", :id, "--hwvirtex", "off"]
    vb.customize ["modifyvm", :id, "--nestedpaging", "off"]
    vb.customize ["modifyvm", :id, "--paravirtprovider", "none"]
  end

  config.vm.provision "shell", inline: <<-SHELL
    apt-get update -y
    apt-get upgrade -y
  SHELL
end
```

Valider :

```
arij@server26:~/vagrant-ubuntu$ vagrant validate
Vagrantfile validated successfully.
```

16- Depuis la VM, connectez-vous à la VM créée par Vagrant avec : *vagrant ssh* README.md : ajoutez une capture d'écran et comparez cette méthode avec la création manuelle de la VM.

a. Tenter vagrant ssh

```
═══════════════════════════════════════
   Q16 — Connexion SSH à la VM Vagrant
═══════════════════════════════════════

=== 1. État de la VM Vagrant ===
Current machine states:

default                   not created (virtualbox)

The environment has not yet been created. Run `vagrant up` to
create the environment. If a machine is not created, only the
default provider will be shown. So if a provider is not listed,
then the machine is not created for that environment.

=== 2. Tentative de connexion SSH ===
VM must be created before running this command. Run `vagrant up` first.

=== 3. Rappel de la limitation ===
INFO: Your CPU does not support KVM extensions
INFO: For more detailed results, you should run this as root
HINT:   sudo /usr/sbin/kvm-ok
arij@server26:~/vagrant-ubuntu$
```

17- Modifiez le Vagrantfile pour configurer automatiquement :le nom de la VM

l'adresse IP privée, la mémoire et le nombre de CPU

README.md : ajoutez la configuration utilisée et le résultat de vagrant status.

a. Modifier le Vagrantfile :

```
 GNU nano 8.7.1                                          Vagrantfile *

Vagrant.configure("2") do |config|

  # -----------------------------
  # Box Ubuntu Server
  # -----------------------------
  config.vm.box = "generic/ubuntu2204"

  # -----------------------------
  # 1. Nom de la VM (hostname)
  # -----------------------------
  config.vm.hostname = "vagrant-ubuntu"

  # -----------------------------
  # 2. Adresse IP privée
  # -----------------------------
  # Réseau privé accessible depuis la VM hôte uniquement
  config.vm.network "private_network", ip: "192.168.56.10"

  # -----------------------------
  # 3. Provider VirtualBox
  # -----------------------------
  config.vm.provider "virtualbox" do |vb|

    # Nom affiché dans VirtualBox
    vb.name = "vagrant-ubuntu"

    # 3. Mémoire RAM allouée (en Mo)
    vb.memory = 1024
```

b. Valider le Vagrantfile :

```
arij@server26:~/vagrant-ubuntu$ vagrant validate
Vagrantfile validated successfully.
```

c. Tenter vagrant status :

```
arij@server26:~/vagrant-ubuntu$ vagrant status
Current machine states:

default                   not created (virtualbox)

The environment has not yet been created. Run `vagrant up` to
create the environment. If a machine is not created, only the
default provider will be shown. So if a provider is not listed,
then the machine is not created for that environment.
arij@server26:~/vagrant-ubuntu$
```

Le contenu du Vagrantfile

```
   Q17 — Vagrantfile configuré

# -*- mode: ruby -*-
# vi: set ft=ruby :

# ============================================================
# Vagrantfile — VM Ubuntu Server (Question 17)
# Auteur : Arij Belaid
# ============================================================
# Configuration automatique :
#   - Nom de la VM       : vagrant-ubuntu
#   - Adresse IP privée  : 192.168.56.10
#   - Mémoire RAM        : 1024 Mo (1 Go)
#   - Nombre de CPU      : 2 vCPU
# ============================================================

Vagrant.configure("2") do |config|

  # -----------------------------
  # Box Ubuntu Server
  # -----------------------------
  config.vm.box = "generic/ubuntu2204"

  # -----------------------------
  # 1. Nom de la VM (hostname)
  # -----------------------------
  config.vm.hostname = "vagrant-ubuntu"

  # -----------------------------
  # 2. Adresse IP privée
  # -----------------------------
```

```
# ------------------------------
# 2. Adresse IP privée
# ------------------------------
# Réseau privé accessible depuis la VM hôte uniquement
config.vm.network "private_network", ip: "192.168.56.10"

# ------------------------------
# 3. Provider VirtualBox
# ------------------------------
config.vm.provider "virtualbox" do |vb|

  # Nom affiché dans VirtualBox
  vb.name = "vagrant-ubuntu"

  # 3. Mémoire RAM allouée (en Mo)
  vb.memory = 1024

  # 4. Nombre de CPU (vCPU)
  vb.cpus = 2

  # Désactivation de la virtualisation imbriquée (limitation VT-x de l'hôte)
  vb.customize ["modifyvm", :id, "--hwvirtex", "off"]
  vb.customize ["modifyvm", :id, "--nestedpaging", "off"]
  vb.customize ["modifyvm", :id, "--paravirtprovider", "none"]
end

# ------------------------------
# Provisionnement : mise à jour
# ------------------------------
config.vm.provision "shell", inline: <<-SHELL
  apt-get update -y
  apt-get upgrade -y
SHELL
```

Validation + Statut :

```
═══════════════════════════════════════════
   Q17 — Validation et état
═══════════════════════════════════════════

=== vagrant validate ===
Vagrantfile validated successfully.

=== vagrant status ===
Current machine states:

default                   not created (virtualbox)

The environment has not yet been created. Run `vagrant up` to
create the environment. If a machine is not created, only the
default provider will be shown. So if a provider is not listed,
then the machine is not created for that environment.
arij@server26:~/vagrant-ubuntu$
```

V. Migration du Portfolio vers Next.js :

18- Créez une nouvelle version du DevSecOps Portfolio avec Next.js. Conservez au minimum les mêmes sections que dans la version HTML/CSS/JavaScript. README.md : ajoutez la commande de création du projet et une capture de l'application.

a. *Installation de Node.js (version LTS) et npm*

```
arij@server26:~/cv-onepage$ sudo apt install -y nodejs npm
Installation de :
  nodejs  npm

Installation de dépendances :
  alacritty                                node-delegates                node-object-visit
  eslint                                   node-depd                     node-once
  gyp                                      node-diff                     node-opener
  handlebars                               node-doctrine                 node-optimist
  javascript-common                        node-electron-to-chromium     node-optionator
  libabsl20260107                          node-encoding                 node-osenv
  libada-url0-3                            node-enhanced-resolve         node-p-cancelable
  libauthen-sasl-perl                      node-envinfo                  node-p-limit
  libcares2                                node-err-code                 node-p-locate
  libclone-perl                            node-errno                    node-p-map
  libcrypt-urandom-perl                    node-error-ex                 node-parse-json
  libdata-dump-perl                        node-es-abstract              node-pascalcase
  libencode-locale-perl                    node-es-module-lexer          node-path-dirname
  libfile-basedir-perl                     node-es6-error                node-path-exists
  libfile-desktopentry-perl                node-escape-string-regexp     node-path-is-absolute
  libfile-listing-perl                     node-escodegen                node-path-is-inside
  libfile-mimeinfo-perl                    node-eslint-scope             node-path-scurry
  libfont-afm-perl                         node-eslint-utils             node-path-type
  libhtml-form-perl                        node-eslint-visitor-keys      node-picocolors
  libhtml-format-perl                      node-espree                   node-pify
  libhtml-parser-perl                      node-esprima                  node-pkg-dir
  libhtml-tagset-perl                      node-esquery                  node-postcss
  libhtml-tree-perl                        node-esrecurse                node-postcss-modules-extract-imports
  libhttp-cookies-perl                     node-estraverse               node-postcss-modules-values
  libhttp-daemon-perl                      node-esutils                  node-postcss-value-parser
  libhttp-date-perl                        node-events                   node-prelude-ls
```

B .Vérification des versions installées :

```
arij@server26:~/cv-onepage$ node --version
v22.22.1
arij@server26:~/cv-onepage$ npm --version
9.2.0
arij@server26:~/cv-onepage$
```

c. Créer le projet Next.js

```
arij@server26:~$ npx create-next-app@latest devsecops-portfolio-next
Need to install the following packages:
  create-next-app@16.4.0
Ok to proceed? (y) y
✔Would you like to use the recommended Next.js defaults? › Yes, use recommended defaults
Creating a new Next.js app in /home/arij/devsecops-portfolio-next.

Using npm.

Initializing project with template: app-tw


Installing dependencies:
- next
- react
- react-dom

Installing devDependencies:
- @tailwindcss/turbopack
- @types/node
- @types/react
- @types/react-dom
- eslint
- eslint-config-next
- tailwindcss
- typescript

(                    ) ⸩ idealTree:devsecops-portfolio-next: sill idealTree buildDeps
npm WARN deprecated eslint@9.39.5: This version is no longer supported. Please see https://eslint.org/version-support for other optio
ns.

added 370 packages, and audited 371 packages in 5m
```

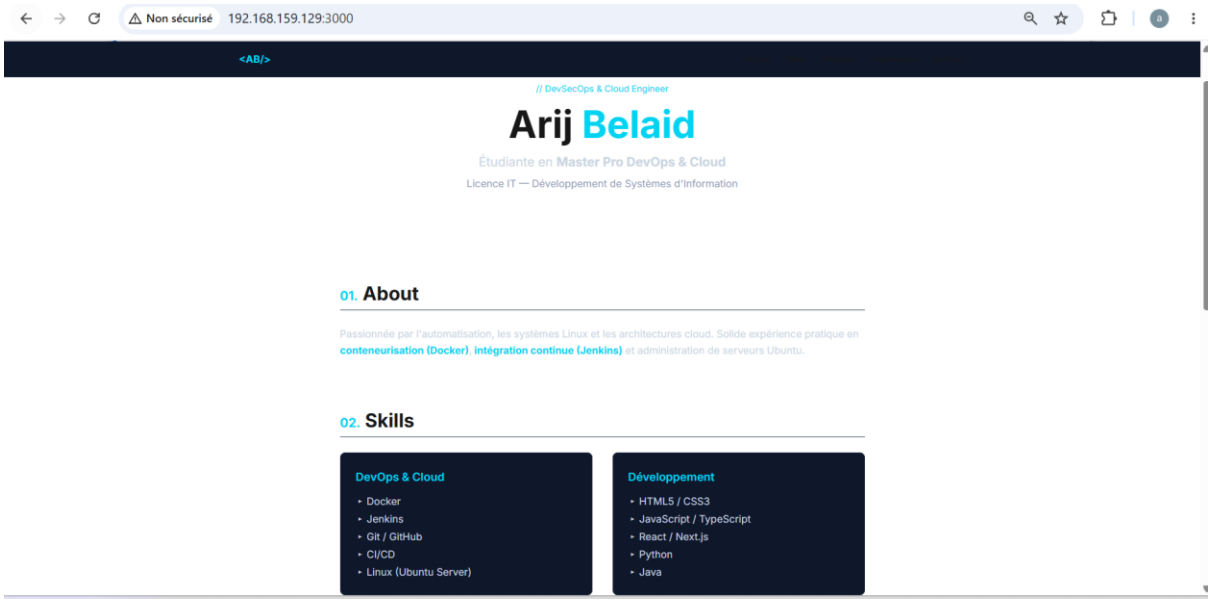
d.vérification :

```
arij@server26:~$ cd devsecops-portfolio-next
arij@server26:~/devsecops-portfolio-next$ ls -la
total 288
drwxrwxr-x   7 arij arij   4096 Oct  7 09:08 .
drwxr-x---  12 arij arij   4096 Oct  7 09:03 ..
drwxrwxr-x   7 arij arij   4096 Oct  7 09:08 .git
-rw-rw-r--   1 arij arij    480 Oct  7 09:03 .gitignore
drwxrwxr-x   3 arij arij   4096 Oct  7 09:08 .next
-rw-rw-r--   1 arij arij    679 Oct  7 09:08 AGENTS.md
-rw-rw-r--   1 arij arij   1450 Oct  7 09:03 README.md
drwxrwxr-x   2 arij arij   4096 Oct  7 09:03 app
-rw-rw-r--   1 arij arij    465 Oct  7 09:03 eslint.config.mjs
-rw-rw-r--   1 arij arij    288 Oct  7 09:08 next-env.d.ts
-rw-rw-r--   1 arij arij    366 Oct  7 09:03 next.config.ts
drwxrwxr-x 294 arij arij  12288 Oct  7 09:08 node_modules
-rw-rw-r--   1 arij arij 223139 Oct  7 09:08 package-lock.json
-rw-rw-r--   1 arij arij    548 Oct  7 09:03 package.json
drwxrwxr-x   2 arij arij   4096 Oct  7 09:03 public
-rw-rw-r--   1 arij arij    666 Oct  7 09:03 tsconfig.json
arij@server26:~/devsecops-portfolio-next$
```

e. Personnaliser le layout

```
arij@server26:~/devsecops-portfolio-next$ nano app/layout.tsx
arij@server26:~/devsecops-portfolio-next$ nano app/page.tsx
```

f. Lancer le serveur de développement



i.push :

```
arij@server26:~/cv-onepage$ git push
Enumerating objects: 31, done.
Counting objects: 100% (30/30), done.
Delta compression using up to 2 threads
Compressing objects: 100% (27/27), done.
Writing objects: 100% (27/27), 149.67 KiB | 940.00 KiB/s, done.
Total 27 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 2 local objects.
To github.com:arijbelaid/cv-onepage.git
   4b58c83..121a5ff  main -> main
```

19- Transformez les principales parties du Portfolio en composants réutilisables : Header, About, Skills, Projects, Contact, Footer
README.md : ajoutez la structure des composants.

a.Créer le dossier components + vérifier

```
arij@server26:~/devsecops-portfolio-next$ mkdir -p ~/devsecops-portfolio-next/components
arij@server26:~/devsecops-portfolio-next$ ls -la ~/devsecops-portfolio-next/
total 296
drwxrwxr-x   8 arij arij   4096 Oct  8 21:00 .
drwxr-x---  12 arij arij   4096 Oct  7 09:03 ..
drwxrwxr-x   7 arij arij   4096 Oct  7 09:08 .git
-rw-rw-r--   1 arij arij    480 Oct  7 09:03 .gitignore
drwxrwxr-x   3 arij arij   4096 Oct  7 10:02 .next
-rw-rw-r--   1 arij arij    679 Oct  7 09:08 AGENTS.md
-rw-rw-r--   1 arij arij   1450 Oct  7 09:03 README.md
drwxrwxr-x   2 arij arij   4096 Oct  7 09:21 app
drwxrwxr-x   2 arij arij   4096 Oct  8 21:00 components
-rw-rw-r--   1 arij arij    465 Oct  7 09:03 eslint.config.mjs
-rw-rw-r--   1 arij arij    296 Oct  7 09:22 next-env.d.ts
-rw-rw-r--   1 arij arij    133 Oct  7 09:56 next.config.ts
drwxrwxr-x 294 arij arij  12288 Oct  7 10:01 node_modules
-rw-rw-r--   1 arij arij 223691 Oct  7 10:01 package-lock.json
-rw-rw-r--   1 arij arij    586 Oct  7 10:01 package.json
-rw-rw-r--   1 arij arij     81 Oct  7 10:02 postcss.config.mjs
drwxrwxr-x   2 arij arij   4096 Oct  7 09:03 public
-rw-rw-r--   1 arij arij    666 Oct  7 09:03 tsconfig.json
```

b. Créer les 6 composants

```
arij@server26:~/devsecops-portfolio-next$ nano components/Header.tsx
arij@server26:~/devsecops-portfolio-next$ nano components/About.tsx
arij@server26:~/devsecops-portfolio-next$ nano components/Skills.tsx
arij@server26:~/devsecops-portfolio-next$ nano components/Projects.tsx
arij@server26:~/devsecops-portfolio-next$ nano components/Contact.tsx
arij@server26:~/devsecops-portfolio-next$ nano components/Footer.tsx
```

c. Tester

```
arij@server26:~/devsecops-portfolio-next$ npm run dev

> devsecops-portfolio-next@0.1.0 dev
> next dev

▲ Next.js 16.4.0 (Turbopack)
- Local:         http://localhost:3000
- Network:       http://192.168.159.129:3000
✓ Ready in 2.9s
✓ Running next.config.ts took 690ms

○ Compiling / ...
 GET / 200 in 2.3s (next.js: 1318ms, application-code: 978ms)
```

// DevSecOps & Cloud Engineer

Arij Belaid

Étudiante en **Master Pro DevOps & Cloud**

Licence IT — Développement de Systèmes d'Information

01. About

Passionnée par l'automatisation, les systèmes Linux et les architectures cloud. Solide expérience pratique en **conteneurisation (Docker)**, **intégration continue (Jenkins)** et administration de serveurs Ubuntu.

02. Skills

DevOps & Cloud

- Docker
- Jenkins
- Git / GitHub
- CI/CD
- Linux (Ubuntu Server)

Développement

- HTML5 / CSS3
- JavaScript / TypeScript
- React / Next.js
- Python
- Java

03. Projects

03. Projects

Chaîne CI/CD complète sur Ubuntu Server

Installation et durcissement d'un serveur Ubuntu Server 26.04, déploiement de Docker et Jenkins.

DevSecOps Portfolio (ce site)

Application one-page responsive développée en Next.js 16 et Tailwind CSS.

Application web de gestion

Développement d'une application de gestion avec base de données MySQL (PHP/JS).

04. Experience

Projet académique — DevOps & Cloud

2026 — en cours | Master Pro DevOps & Cloud

- Mise en place d'une chaîne CI/CD complète (Jenkins + Docker)
- Durcissement d'un serveur Ubuntu (SSH, UFW)

Licence IT — Développement de Systèmes d'Information

2023 — 2026 | Formation universitaire

05. Contact

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GITHUB
@arijbelaid

© 2026 Arij Belaid — DevSecOps Portfolio (Next.js)

```
═══════════════════════════════════════
   Q19 — Structure des composants
═══════════════════════════════════════

=== Dossier components/ ===
total 32
drwxrwxr-x 2 arij arij 4096 Oct  8 21:11 .
drwxrwxr-x 8 arij arij 4096 Oct  8 21:00 ..
-rw-rw-r-- 1 arij arij  692 Oct  8 21:10 About.tsx
-rw-rw-r-- 1 arij arij  975 Oct  8 21:11 Contact.tsx
-rw-rw-r-- 1 arij arij  230 Oct  8 21:11 Footer.tsx
-rw-rw-r-- 1 arij arij  791 Oct  8 21:10 Header.tsx
-rw-rw-r-- 1 arij arij 1239 Oct  8 21:11 Projects.tsx
-rw-rw-r-- 1 arij arij 1082 Oct  8 21:11 Skills.tsx

=== Arborescence complète ===
./app/layout.tsx
./app/page.tsx
./components/About.tsx
./components/Contact.tsx
./components/Footer.tsx
./components/Header.tsx
./components/Projects.tsx
./components/Skills.tsx

=== app/layout.tsx (imports) ===
import type { Metadata } from "next";
import { Inter } from "next/font/google";
import "./globals.css";

import Header from "@/components/Header";
import Footer from "@/components/Footer";
```

```
const inter = Inter({ subsets: ["latin"] });

export const metadata: Metadata = {

=== app/page.tsx (imports) ===
import About from "@/components/About";
import Skills from "@/components/Skills";
import Projects from "@/components/Projects";
import Contact from "@/components/Contact";

export default function Home() {
  return (
    <div className="max-w-4xl mx-auto p-8 space-y-20">
      {/* HERO */}
      <section id="hero" className="text-center py-16">
arij@server26:~/devsecops-portfolio-next$
```

20- Déplacez les informations des projets et compétences dans des **fichiers de données ou objets JavaScript séparés**.
README.md : ajoutez un exemple de données utilisées.

a. Créer le dossier data et les fichiers data :

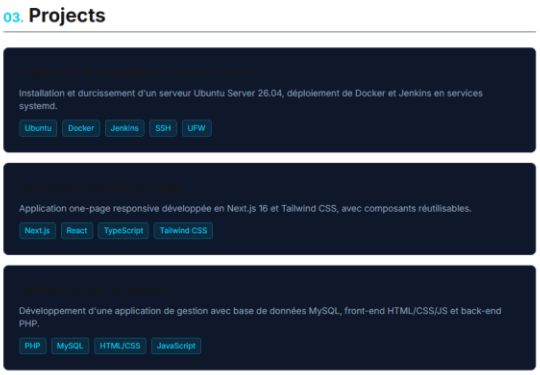
```
arij@server26:~/devsecops-portfolio-next$ nano data/projects.ts
arij@server26:~/devsecops-portfolio-next$ nano data/contacts.ts
arij@server26:~/devsecops-portfolio-next$ nano data/experiences.ts
arij@server26:~/devsecops-portfolio-next$ nano data/index.ts
arij@server26:~/devsecops-portfolio-next$ nano components/Skills.tsx
arij@server26:~/devsecops-portfolio-next$ nano components/Projects.tsx
```

b.vérifier :

```
arij@server26:~/devsecops-portfolio-next$ cd ~/devsecops-portfolio-next
tree -L 2 -I 'node_modules|.next' 2>/dev/null || find . -type f \( -name "*.ts" -o -name "*.tsx" \) | grep -v node_modules | grep -v
.next | sort
./app/layout.tsx
./app/page.tsx
./components/About.tsx
./components/Contact.tsx
./components/Footer.tsx
./components/Header.tsx
./components/Projects.tsx
./components/Skills.tsx
./data/contacts.ts
./data/experiences.ts
./data/index.ts
./data/projects.ts
./data/skills.ts
arij@server26:~/devsecops-portfolio-next$
```

c. Vérifier que page.tsx importe experiencesData :

```
arij@server26:~/devsecops-portfolio-next$ grep -n "experiencesData" app/page.tsx
5:import { experiencesData } from "@/data";
31:      {/* EXPERIENCE — utilise experiencesData */}
37:          {experiencesData.map((exp) => (
```



21- Ajoutez une page dédiée à l'affichage d'un projet du Portfolio :*/projects /projects/project-name :*

a.mettre a jours :

```
arij@server26:~/devsecops-portfolio-next$ nano data/projects.ts
```

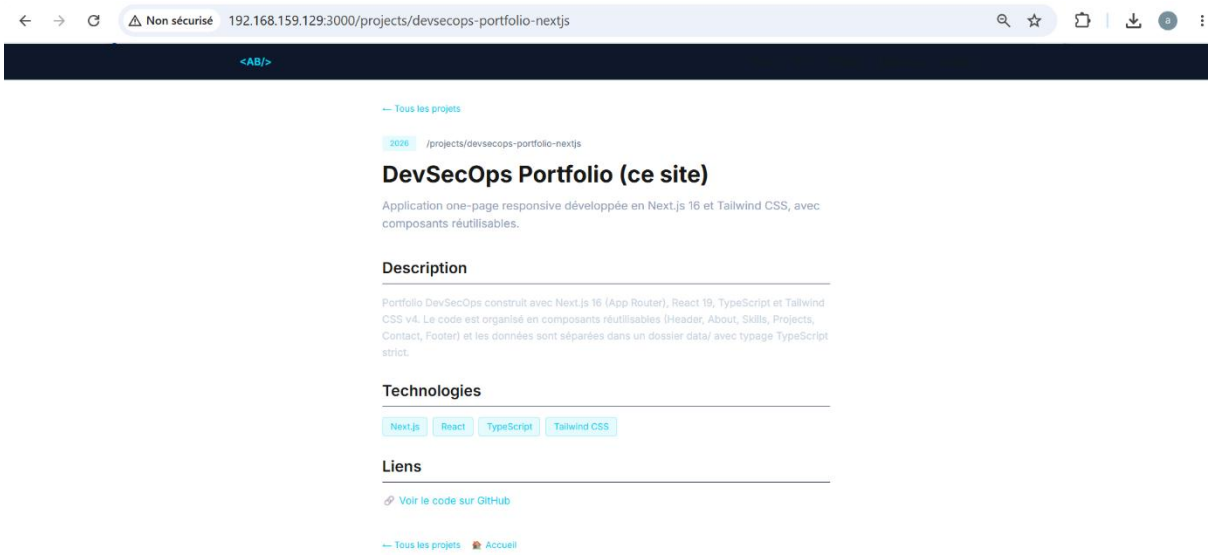
b.Créer la page /projects , page dynamique /projects/[slug]

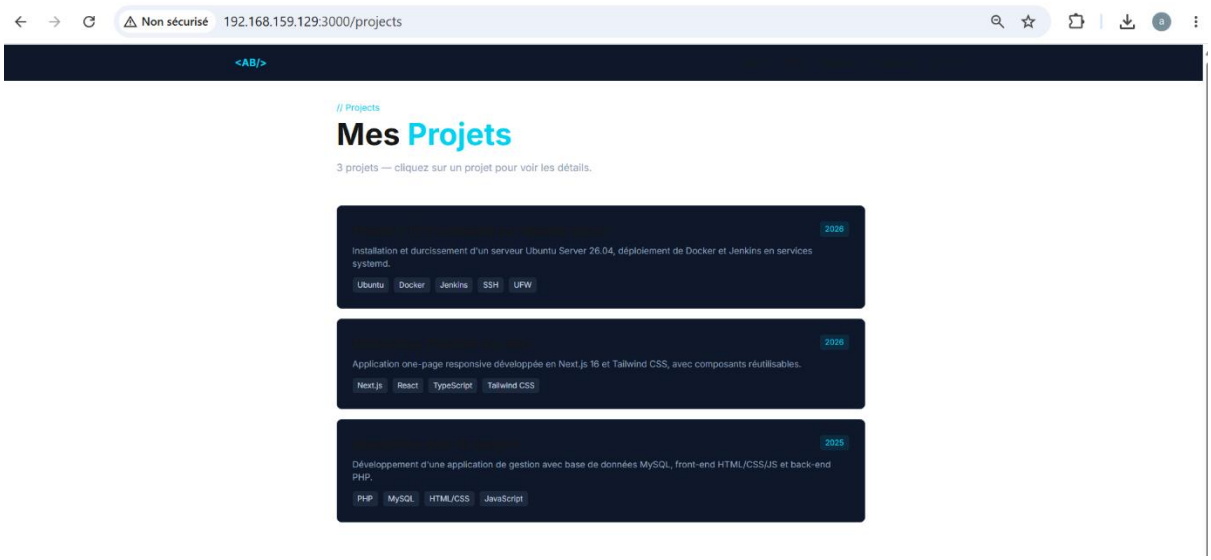
```
arij@server26:~/devsecops-portfolio-next$ mkdir -p app/projects
nano app/projects/page.tsx
arij@server26:~/devsecops-portfolio-next$ mkdir -p "app/projects/[slug]"
arij@server26:~/devsecops-portfolio-next$ nano "app/projects/[slug]/page.tsx"
arij@server26:~/devsecops-portfolio-next$
```

c.Mettre à jour le Header (lien vers /projects) et Mettre à jour components/Projects.tsx (lien vers /projects/[slug])

```
arij@server26:~/devsecops-portfolio-next$ nano components/Header.tsx
arij@server26:~/devsecops-portfolio-next$ nano components/Projects.tsx
```

Exécution :







22- Créez un premier Jenkins Pipeline capable de récupérer automatiquement le dépôt GitHub du DevSecOps Portfolio.
README.md : ajoutez le Jenkinsfile et une capture du pipeline exécuté.

a. Créer le Jenkinsfile +vérification :

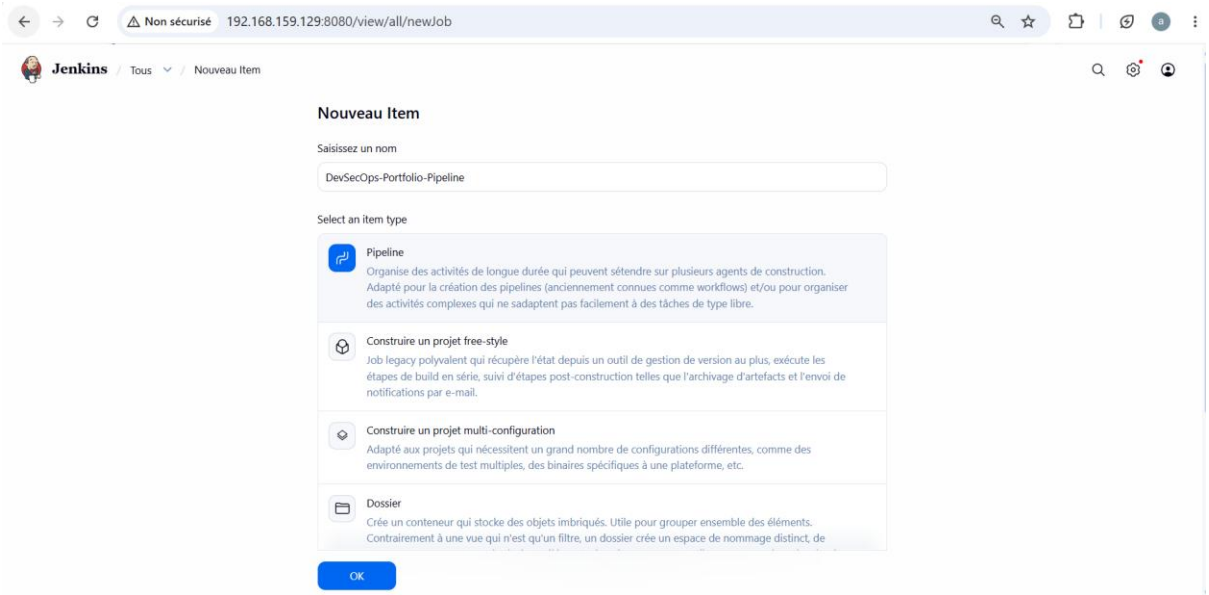
```
arij@server26:~/cv-onepage$ cat Jenkinsfile
pipeline {
    agent any

    stages {
        stage('Checkout') {
            steps {
                echo '🔄 Récupération du dépôt GitHub...'
                checkout scm
                script {
                    def branch = sh(script: "git rev-parse --abbrev-ref HEAD", returnStdout: true).trim()
                    def commit = sh(script: "git rev-parse --short HEAD", returnStdout: true).trim()
                    echo "🌿 Branche : ${branch}"
                    echo "📌 Commit : ${commit}"
                }
            }
        }

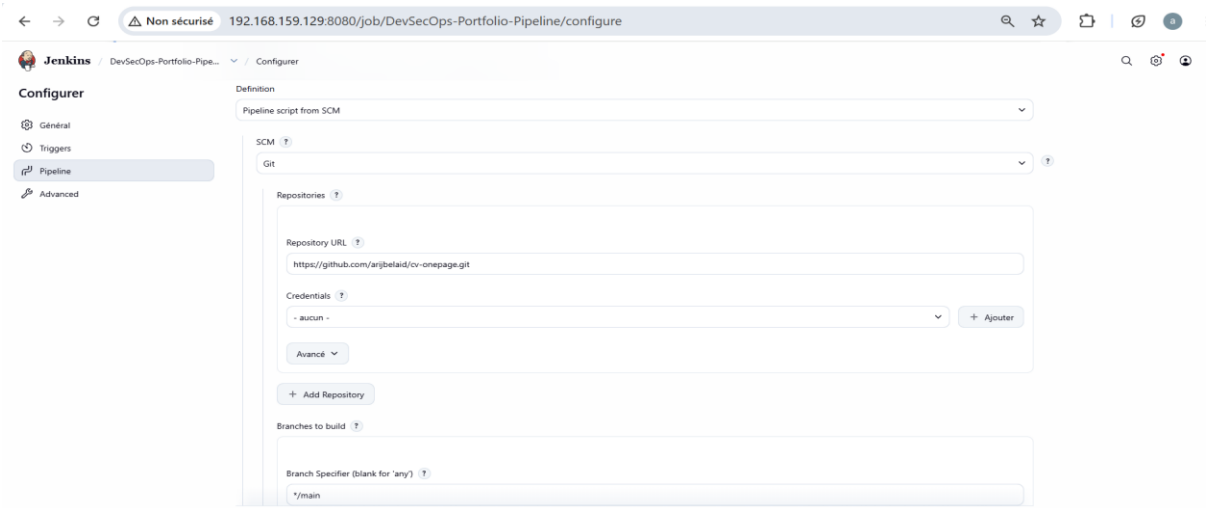
        stage('Verify') {
            steps {
                echo '✅ Vérification des fichiers du projet...'
                sh 'ls -la'
                sh 'test -f index.html && echo "✓ index.html trouvé"'
                sh 'test -f nextjs-portfolio/package.json && echo "✓ nextjs-portfolio/package.json trouvé"'
            }
        }
    }

    post {
        success {
            echo '✅ Pipeline terminé avec succès !'
        }
        failure {
```

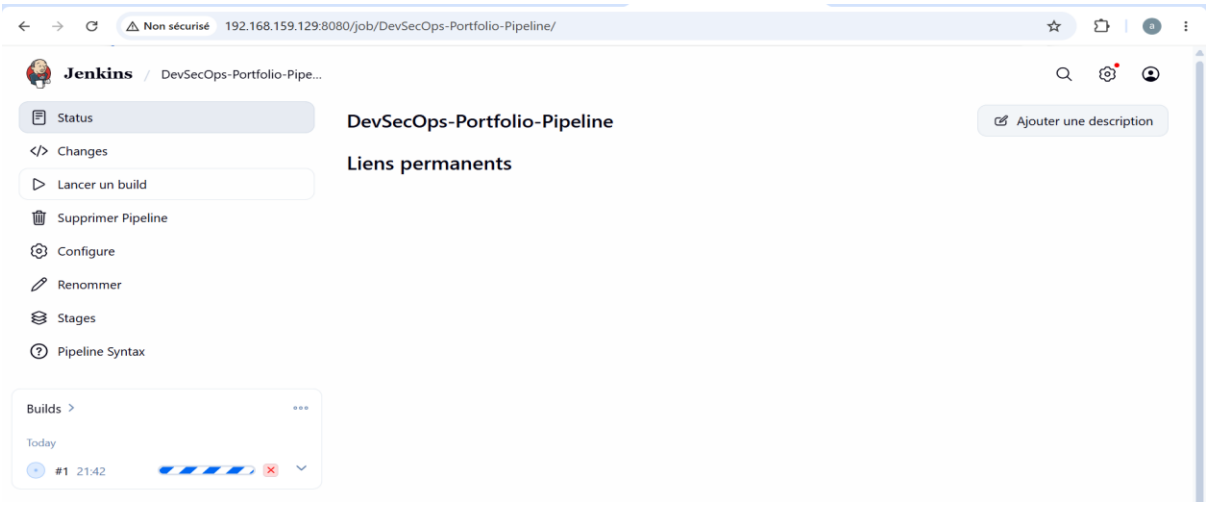
b. Créer le Pipeline dans Jenkins

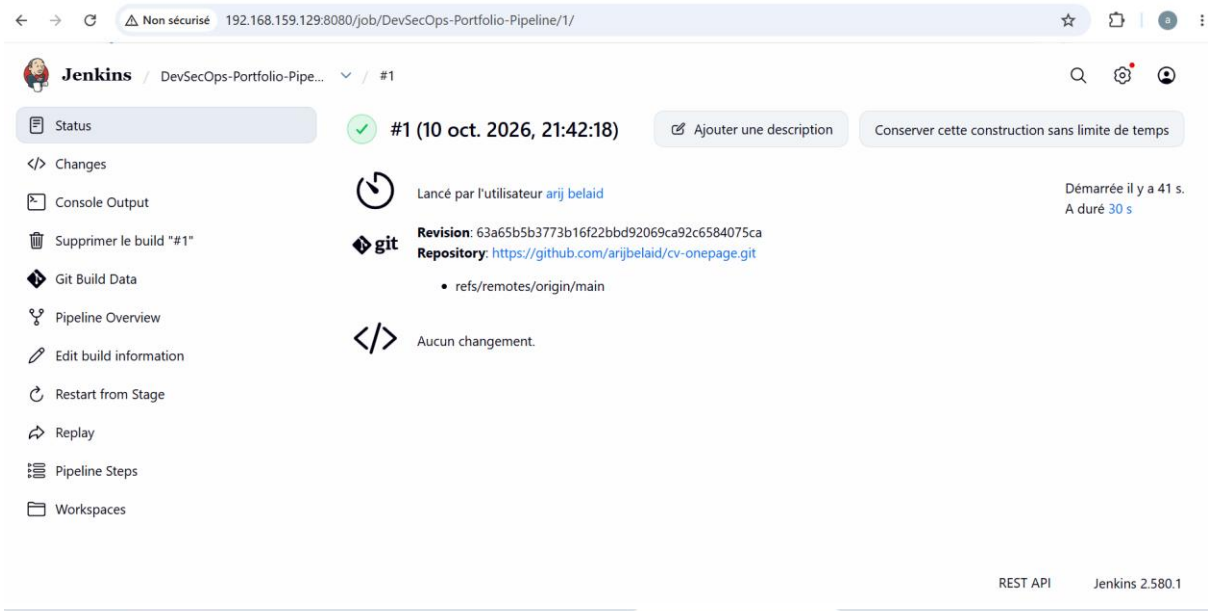


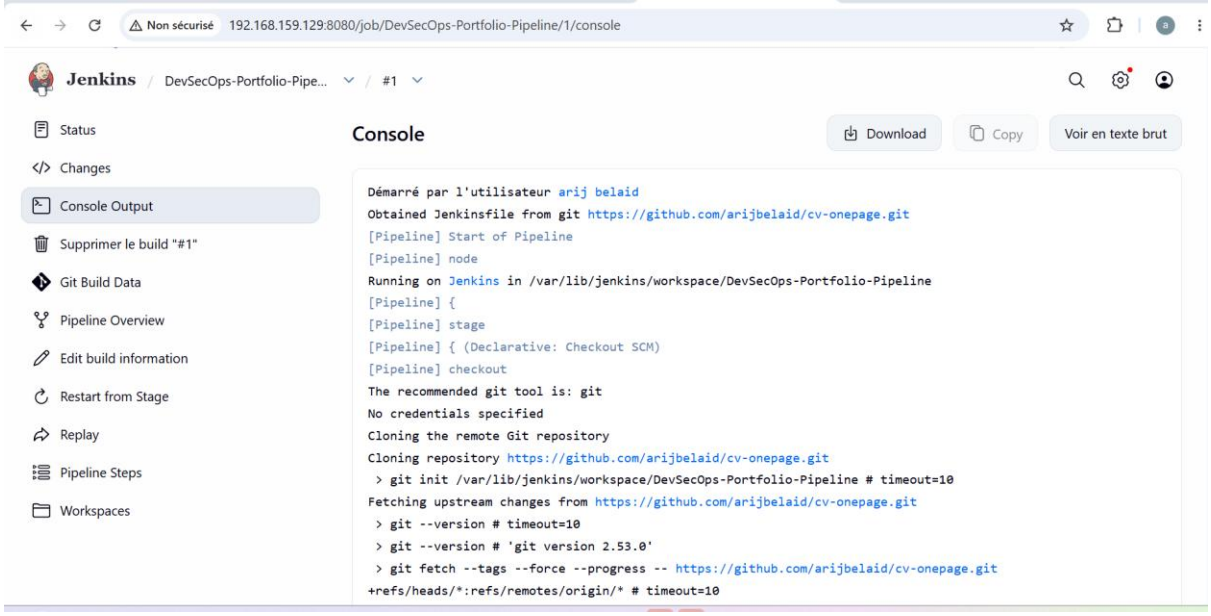
c.Configurer le Pipeline :

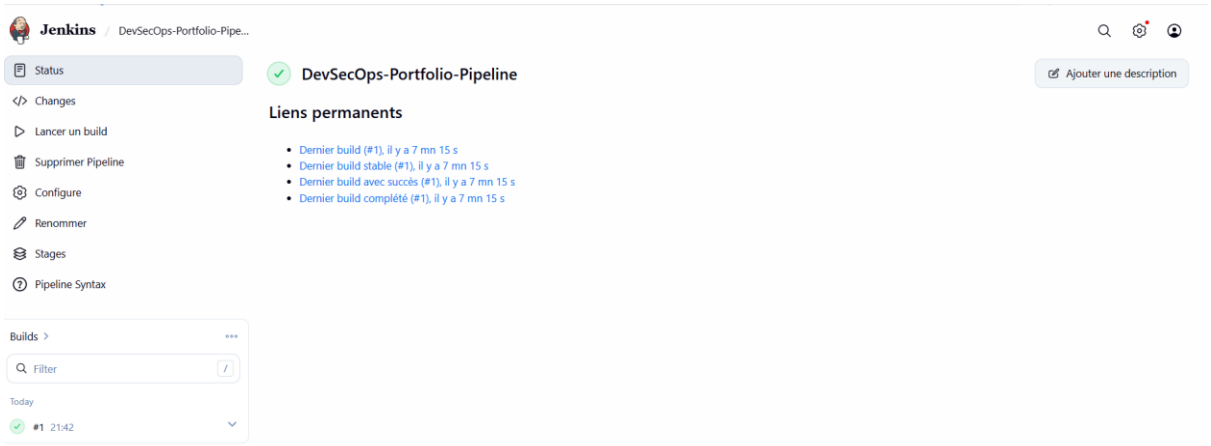


d. Lancer le Pipeline




Non sécurisé
192.168.159.129:8080/job/DevSecOps-Portfolio-Pipeline/1/
Jenkins
DevSecOps-Portfolio-Pipe...
#1
Status
Changes
Console Output
Supprimer le build "#1"
Git Build Data
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
#1 (10 oct. 2026, 21:42:18)
Ajouter une description
Conserver cette construction sans limite de temps
Lancé par l'utilisateur arij belaid
Démarrée il y a 41 s.
A duré 30 s
git
Revision: 63a65b5b3773b16f22bbd92069ca92c6584075ca
Repository: https://github.com/arijbelaid/cv-onepage.git
refs/remotes/origin/main
Aucun changement.
REST API
Jenkins 2.580.1


Non sécurisé
192.168.159.129:8080/job/DevSecOps-Portfolio-Pipeline/1/console
Jenkins
DevSecOps-Portfolio-Pipe...
#1
Status
Changes
Console Output
Supprimer le build "#1"
Git Build Data
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Console
Download
Copy
Voir en texte brut
Démarré par l'utilisateur arij belaid
Obtained Jenkinsfile from git https://github.com/arijbelaid/cv-onepage.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/DevSecOps-Portfolio-Pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
The recommended git tool is: git
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/arijbelaid/cv-onepage.git
> git init /var/lib/jenkins/workspace/DevSecOps-Portfolio-Pipeline # timeout=10
Fetching upstream changes from https://github.com/arijbelaid/cv-onepage.git
> git --version # timeout=10
> git --version # 'git version 2.53.0'
> git fetch --tags --force --progress -- https://github.com/arijbelaid/cv-onepage.git
+refs/heads/*:refs/remotes/origin/* # timeout=10

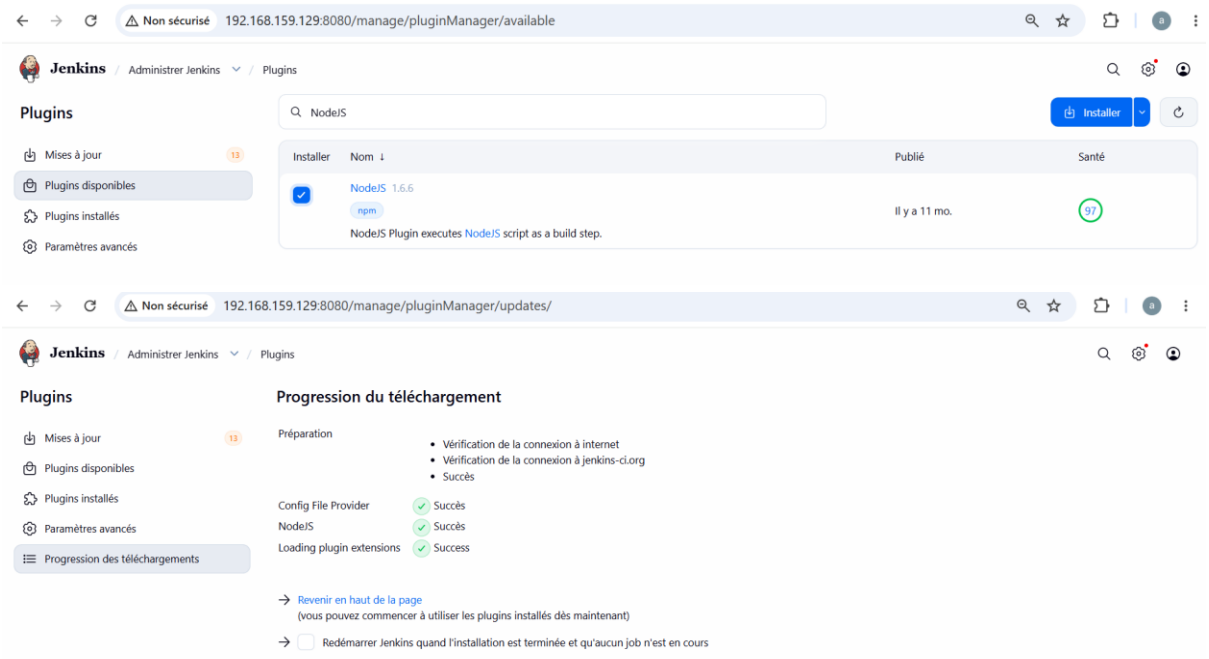

Jenkins
DevSecOps-Portfolio-Pipe...
Status
Changes
Lancer un build
Supprimer Pipeline
Configure
Renommer
Stages
Pipeline Syntax
Builds
Filter
Today
#1 21:42
DevSecOps-Portfolio-Pipeline
Ajouter une description
Liens permanents
Dernier build (#1), il y a 7 mn 15 s
Dernier build stable (#1), il y a 7 mn 15 s
Dernier build avec succès (#1), il y a 7 mn 15 s
Dernier build complété (#1), il y a 7 mn 15 s

23- Ajoutez au pipeline une étape permettant d'installer les dépendances du projet Next.js.
README.md : ajoutez une capture du stage correspondant

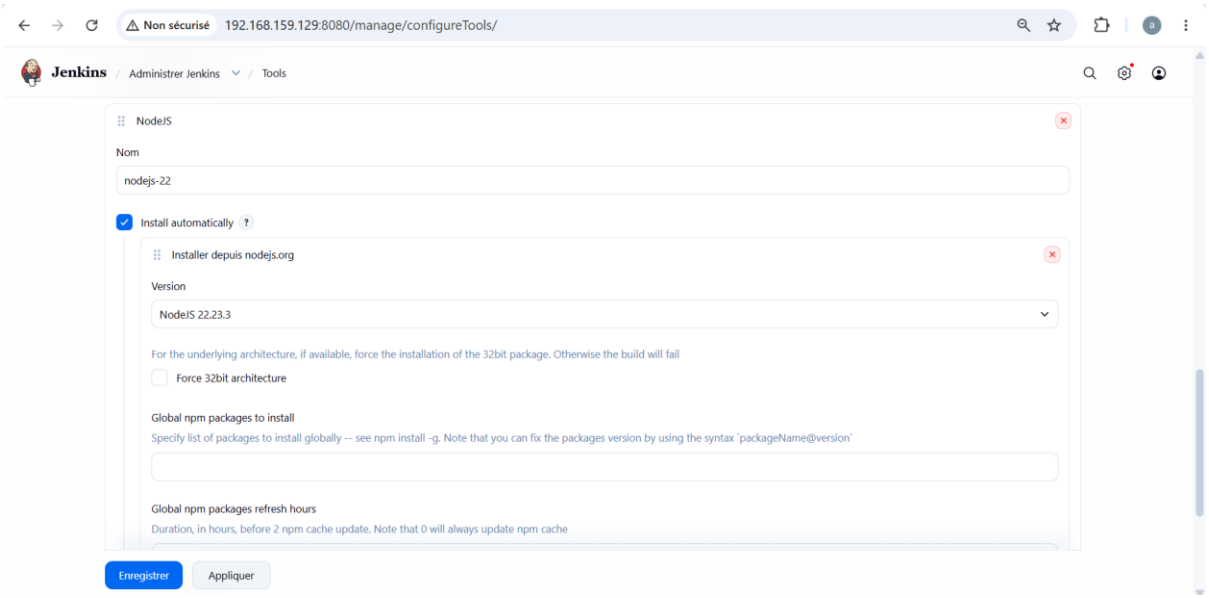
a. Vérifier Node.js sur la VM

```
arij@server26:~/cv-onepage$ node --version
npm --version
v22.22.1
9.2.0
arij@server26:~/cv-onepage$
```

b. Installer le plugin NodeJS dans Jenkins



c. Configurer Node.js dans les outils Jenkins

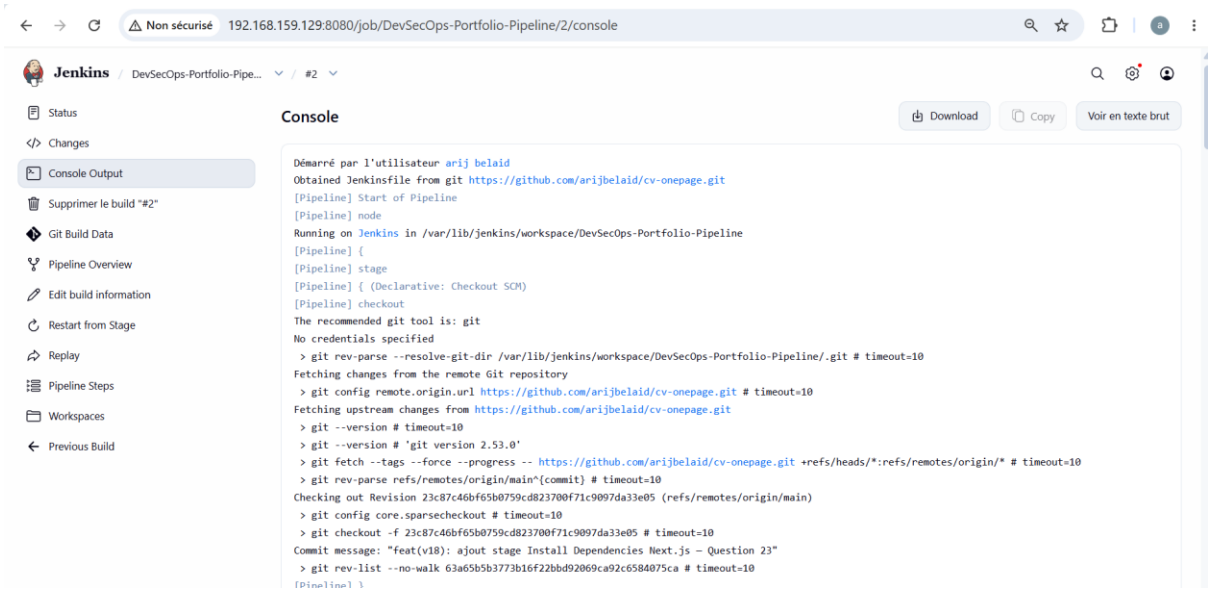


d.modifier le Jenkinsfile

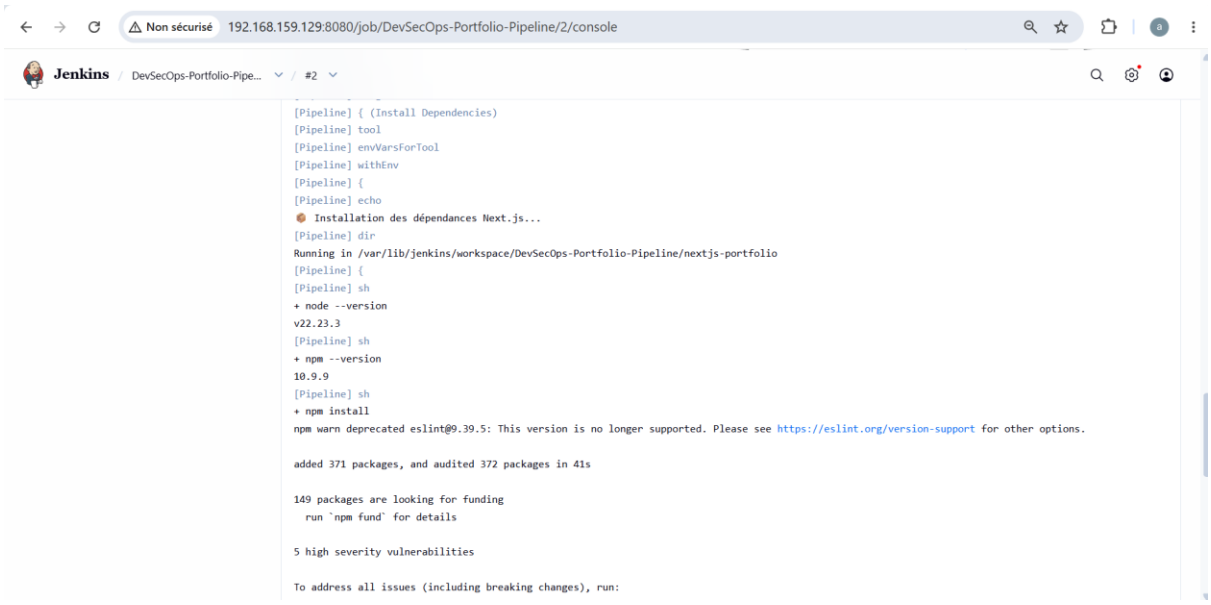
e. Pousser sur GitHub

```
arij@server26:~/cv-onepage$ git add Jenkinsfile
arij@server26:~/cv-onepage$ git commit -m "feat(v18): ajout stage Install Dependencies Next.js – Question 23"
[main 23c87c4] feat(v18): ajout stage Install Dependencies Next.js – Question 23
 1 file changed, 16 insertions(+)
arij@server26:~/cv-onepage$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 530 bytes | 265.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:arijbelaid/cv-onepage.git
   63a65b5..23c87c4  main -> main
```

f. Relancer le pipeline



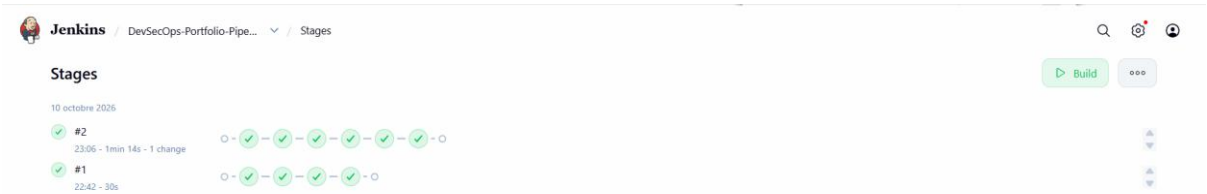
Stage Install Dependencies



Fin du pipeline

```
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
✅ Pipeline terminé avec succès !
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Stage View



24- Ajoutez au pipeline une étape permettant de construire automatiquement l'application Next.js.

README.md : ajoutez une capture du résultat du build.

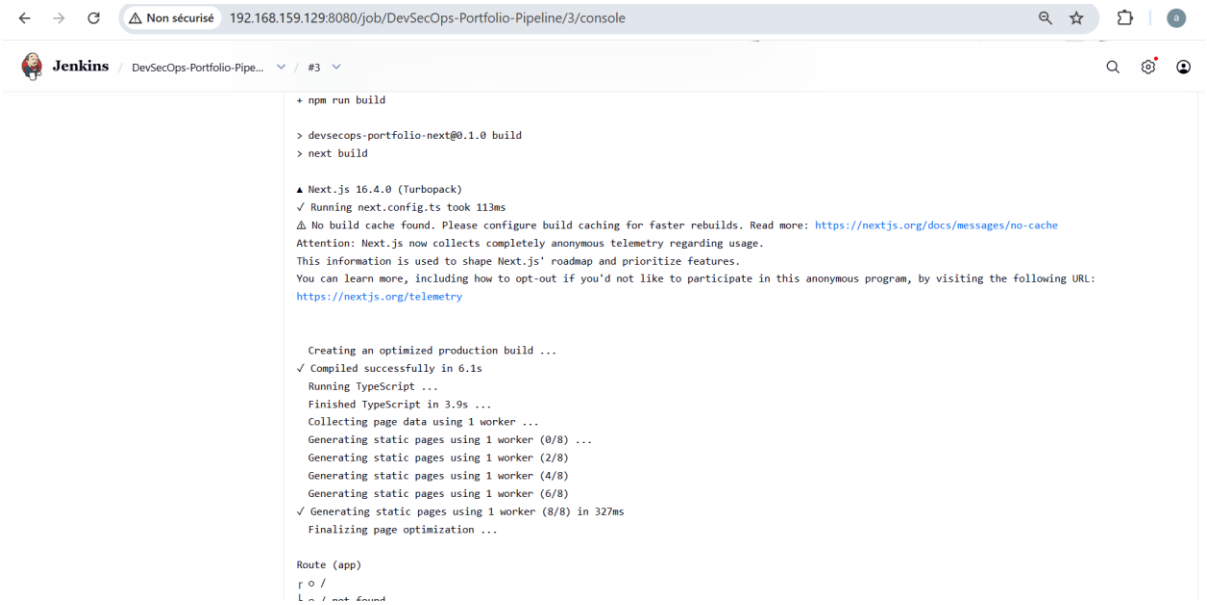
a. Modifier le Jenkinsfile + verification

```
arij@server26:~/cv-onepage$ cd ~/cv-onepage
nano Jenkinsfile
arij@server26:~/cv-onepage$ cat Jenkinsfile | grep -A 8 "stage('Build')"
        stage('Build') {
            steps {
                echo '🏗️ Construction de l\'application Next.js...'
                dir('nextjs-portfolio') {
                    sh 'npm run build'
                    sh 'ls -la .next/ | head -10'
                }
            }
        }
arij@server26:~/cv-onepage$
```

b. Pousser sur GitHub

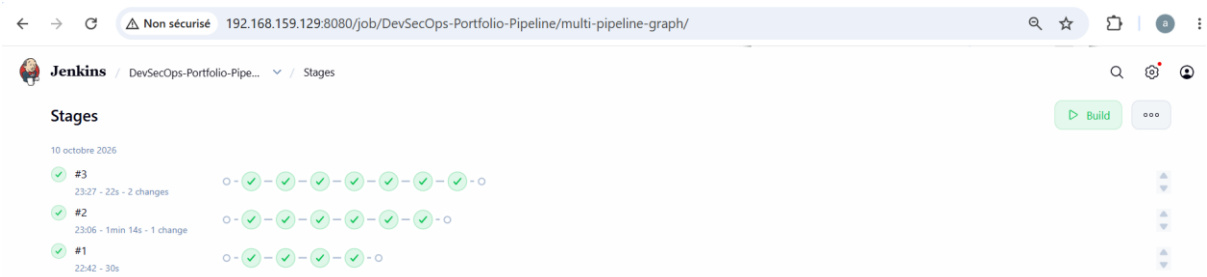
```
arij@server26:~/cv-onepage$ git add Jenkinsfile
git commit -m "feat(v19): ajout stage Build Next.js — Question 24"
git push
[main 3b6c5a9] feat(v19): ajout stage Build Next.js — Question 24
 1 file changed, 10 insertions(+), 1 deletion(-)
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 412 bytes | 412.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:arijbelaid/cv-onepage.git
   1a375a4..3b6c5a9  main -> main
arij@server26:~/cv-onepage$
```

c. Relancer le pipeline





Stage View



d.Commit + Push sur GitHub

```
arij@server26:~/cv-onepage$ cd ~/cv-onepage
git add .
git status
git commit -m "docs: ajout captures Q24 (Jenkins Build Next.js)"
git push
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   captures/39-jenkins-pipeline-stages-v3.png
        new file:   captures/40-jenkins-build-console.png
        new file:   captures/41-jenkins-build-success.png

[main f217401] docs: ajout captures Q24 (Jenkins Build Next.js)
 3 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 captures/39-jenkins-pipeline-stages-v3.png
 create mode 100644 captures/40-jenkins-build-console.png
 create mode 100644 captures/41-jenkins-build-success.png
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 195.16 KiB | 1.48 MiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:arijbelaid/cv-onepage.git
   3b6c5a9..f217401  main -> main
arij@server26:~/cv-onepage$
```